

# The Fast Learning Brain

Natural-language goals in; a playlist that adapts within the session and forgets at the boundary. **A design, implementation, and verification record** for SynAmp’s smart playlists — with measurement honesty and cultural guardrails built into the machinery, not bolted on as a disclaimer.

| PROJECT                                                       | SLICE                                                                   | REPOSITORY                                               | PREPARED                     | EVIDENCE                                                        |
|---------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------|------------------------------|-----------------------------------------------------------------|
| SynAmp – self-hosted music system                             | Intent layer · epoch-v1 learning · sequencing · endpoints · web readout | ~/_dev/web-apps/SynAmp · branch feat/fast-learning-brain | 2026-10-07 · America/Chicago | DELIVERY/verification/ · tools/brain-lab/ · evidence/final-lab/ |
| STATUS                                                        |                                                                         |                                                          |                              |                                                                 |
| Verified slice · reviews closed · frozen manifest at delivery |                                                                         |                                                          |                              |                                                                 |

One commit on `feat/fast-learning-brain` · all magnitudes are candidate values pending real-listening tuning · fixtures are synthetic and are not evidence about real music

# Executive summary

**What was asked, and what exists.** SynAmp's fast-learning brain was asked to do this: state a goal in plain words ("pump me up", "I need to focus") and get a smart playlist that learns fast from listening and forgets at the session boundary. That slice now exists: branch `feat/fast-learning-brain`, a single commit; new modules in `intent/` and `learning/`; the sequencing module `query/sequence.ts`; new `/api/v1/brain` routes (session readout, forget, proposals); and a web "This session" panel.

**The two halves.** The translation layer carries nine goal bundles (focus, pump\_up, dance, calm, sleep, catharsis, nostalgia, drive, chores), whose research weights never outrank the user's own words; accuracy classes that include contradictions — resolved as two labelled readings plus a clarifying ask, never a silent choice; asks, never guesses for vague or impossible requests; and sequencing arcs that re-order the filtered list without ever betraying a hard constraint. The learning layer is epoch-v1: sessions split on a 30-minute silence gap, a day change, or a session-key change, with a 30-minute live window; evidence and hides compose through one bounded saturation that can re-order but never breaks a hard rule; reliability-weighted keep/skip centroids learn small attribute-level tastes; proposals are the only channel that crosses a session; exploration ships off.

**The guarantees that answer the original ask.** *Isolation*: a different day, session, or hour cannot bend this moment's ranking — proven by byte-equality when foreign events are appended. *Honesty*: null is never zero, and a method's rejection is not musical truth. *Consent*: nothing crosses a session without a yes, and forget is one button — append-only; the log is marked, never edited.

**Verification posture.** The suite stands at 218 tests / 216 pass, plus one pre-existing macOS red and one skip; both type-checks exit 0; a deterministic 8-scenario / 86-check harness re-runs byte-identically apart from its timestamp. Independent reviews (code, bias, psychology, production) closed with fixes landed, and every evidence path is carried in the delivery manifest. Honest limits: magnitudes and thresholds are candidate values — hypotheses, not tuned evidence; fixtures are synthetic, so no real-music evidence yet.

## Where to go next

- Run one real-library session (read `GET /api/v1/brain/session`), then a tuning pass.
- Land the P4 producers before declared fields (voice, instruments, mood) become real.
- Revisit the conservative proposal thresholds once real data exists.

Chapter 1 explains how words become rules; Chapter 2 how learning stays in-session; Chapter 3 how measurement stays honest across cultures; Chapter 4 what was verified and how to roll back.

# Words in, music out — the translation layer

## The problem

SynAmp's measurement layer speaks in numbers — bpm, onset rate, percussiveness, integrated loudness — while people speak in intentions: "I need to focus", "pump me up to win this game", "I want to dance". The vocabularies do not map cleanly: a request can be accurate, partial, vague, contradictory, or impossible with today's signals ("90s Mongolian throat singing only"). The translation layer's contract: turn words into a candidate plan while never pretending to understand more than it does.

## The pipeline

Text enters `interpretGoal()` (parser `intent-v1`; `apps/brain/src/intent/interpret.ts`), which composes the rule-based draft parser (`draft.ts`) with the goal lexicon (`lexicon.ts`, `goal-lexicon/1`). The draft converts recognised phrases into an *untrusted* plan object; the lexicon merges goal bundles, applies supersessions, and resolves library-relative bands. Every emitted reading is validated by `validatePlan()` — the closed, versioned plan schema (v2.0, SHA-256-hashed); readings that fail are dropped with a note, never shipped (B1 §Conclusion). Evaluation (`evaluate.ts`) filters and scores the union of the hard guard and similarity channels; `sequenceTracks()` re-orders the already-filtered strict list per the arc; the queue is built from that list.

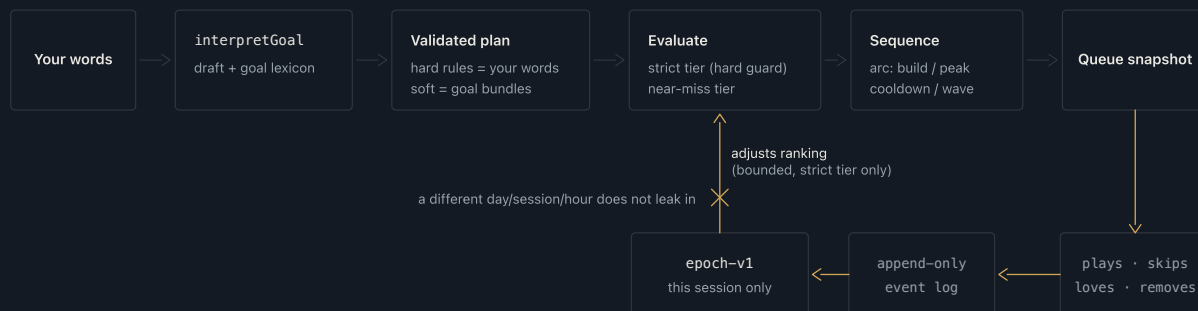


Figure 1 · From your words to your queue — and the loop that learns back, within this session only.

**Hard vs soft — two authorities in one plan.** Hard constraints come from the user's own words: "no words", "no piano", "nothing too slow/relaxing" compile to hard, `explicit_exclusion` rules ("hard; unmeasured fails") — flagged for confirmation, never loosened downstream. Soft constraints come only from research bundles and modifiers; they weight the ranking, never filter. Tests assert that no `goal_*` / `mod_*` term is ever hard (intent.test.ts house rules; B1 §Analysis 4).

## Nine goals, two modifiers

The lexicon carries nine goals — focus, pump\_up, dance, calm, sleep, catharsis, nostalgia, drive, chores — each with word-boundary, negation-guarded trigger phrases ("I need to focus", "pump me up", "I want to dance"), plus two tempo modifiers (fast, slow — soft leans only). A **goal bundle** is the goal's set of soft weighted terms (field/op/band/weight plus mechanism and [MP-n]/[CC-n]

refs) taken from A1 §2, emitted as `goal_<goal>_<n>`. Emitted weights are floor-normalised so each bundle totals  $\leq 1.0$  with every term  $\leq 0.5$ , honouring A1 §2.0's dominance convention — user words and bounded feedback outrank research defaults — while research weights stay on the lexicon entries for traceability (B1 §Analysis 1). Weights are research suggestions, not tuned values (A1 §2.0).

**One reading, not a stack.** When a goal matches, its bundle *supersedes* the draft's proxy constraints for that reading — dropped, never kept twice, audited (flagship: draft `focus_arousal / focus_pulse` superseded). The same rule fixed "I'm driving to work": the draft's generic "energetic" arousal proxy is superseded by the drive bundle (H1-4; C3 F-A1-1). A supersede can never drop a user exclusion (intent.test.ts).

**Contradictions, deliberately.** Conflicts resolve in a documented order — opposing goal pairs (pump\_up↔calm, dance↔sleep), then the documented fast+calm pair, then hard numeric bounds with an empty intersection. Each conflict yields **two readings** plus a clarifying ask; neither side is chosen silently (ties resolve to the first, so the interface must show both labels or a chooser). "fast relaxing bangers" produces exactly that: reading A "energetic but soft" keeps the fast side and adds soft texture terms; reading B "literally calm" drops the fast side and keeps the calm bundle (B1 §Analysis 5; `FAST_CALM_PAIR`).

**Two moods in one ask.** "music for my workout then sleep" is not merged into a self-cancelling wash: the then/after cue builds one reading for the first goal only ("pump up", arc build) plus an ask to request the second phase later (H1-6). Declared conflicts still take precedence ("relax after my workout" yields the two-reading pair).

**Asks, never guesses.** Vague and impossible requests return no readings ( `chosen_index: -1` ) and 1–3 plain-language asks with reasons: "play me something good" is `vague` — surface verbs ("play", "queue", "start") are glue, not content (a harness-caught, regression-pinned fix) — and "not four-on-the-floor" stays `impossible`, the ask naming the phrase (B1 §Evidence).

**Audit transparency.** Every interpretation ships an audit trail (phrase → becomes; kinds: hard/soft/goal/exclusion/unparsed/note), rendered as the expandable **"How I read it (N)"** ( `Describe.tsx:98` ): nothing is inferred silently.

## Sequencing: arcs that cannot betray the ask

`sequenceTracks()` only re-orders the already-filtered strict list — it never filters, adds, or scores — so every hard guard stays true at every position (A1 §3; sequence.ts header). Arcs: build (ascending), cooldown (descending), peak (rise then relax), wave (interleaved; implemented and tested, unused by any goal today). Goal arcs: pump\_up→build, dance→peak, calm/sleep→cooldown, others flat. "Intensity" means exactly the documented *measured pace/energy proxy*: bpm (weight .35;  $\times 0.5$  when tempo confidence is low/missing — octave errors are a known failure class), onset rate (.25), percussiveness (.20), loudness (.20 — deliberately smallest: LUFS  $\neq$  perceived intensity [CC-43.45]). Below 60% coverage the module keeps the ranked order and returns an honest note — "not enough measured pace/energy data (N% coverage) — kept the ranked order"; tracks without measurements keep their positions; null never counts as zero (sequence.ts; H1-2 reworded the note from "intensity" to "pace/energy").

## Worked examples

1. **"I need to focus. no words, no piano, nothing too slow/relaxing."** → `specific`; one reading; the user's words stay hard; focus bundle F1–F10 merges; arc flat.
2. **"pump me up to do this task / win this game"** → `goal_pump_up_*`; arc build.
3. **"I want to dance"** → `goal_dance_*`; arc peak.
4. **"play me something good"** → `vague`; no readings; asks carry the next step.
5. **"fast relaxing bangers"** → `contradictory`; two readings ("energetic but soft" / "literally calm") plus one clarifying ask.

All five are receipt-verified spot-checks (B1 §Evidence; intent.test.ts; re-run in H1 §3).

| PHRASE                                                            | OUTCOME             | KEY RULES                                                                                                                                                                      |
|-------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "I need to focus. no words, no piano, nothing too slow/relaxing." | specific, 1 reading | hard <code>no_words</code> / <code>no_piano</code> / <code>not_slow</code> / <code>not_relaxing</code> + <code>goal_focus_1...10</code> ; supersede of draft proxies; arc flat |
| "pump me up... win this game"                                     | specific/partial    | <code>goal_pump_up_*</code> ; arc build                                                                                                                                        |
| "I want to dance"                                                 | specific            | <code>goal_dance_*</code> ; arc peak                                                                                                                                           |
| "play me something good"                                          | vague               | 0 readings; asks (surface verbs = glue)                                                                                                                                        |
| "fast relaxing bangers"                                           | contradictory       | two readings + ask; <code>FAST_CALM_PAIR</code>                                                                                                                                |

All behaviours above are test-verified against **synthetic fixtures** — they prove logic, not classifier accuracy on real music; nothing here is tuned or validated on a real library. At fix time the brain suite stood at **218 tests (216 passing; one pre-existing macOS case-fold red, one environment skip)** — final counts live in verification (ch. 4).

## Bias and honesty notes (quotable)

- **Bands are research starting points, not universals.** "Tempo/energy bands are research starting points, not universal — they stay soft and learn from your session feedback and spot-check corrections." ( `SHARED_ASSUMPTIONS.calibration` ; A1 §2.0; A2 CC-29/30/36/37).
- **Declared-field terms are inert until producers land.** Voice, instrument and mood-style fields have no producer today; terms on them emit `unknown_policy: "neutral"` with the note "no producer yet — inert on real data" and are unknown for every track on a real library. (C3 F-D-1 precision: they don't change base ordering; the magnitude dilution is a tracked claim item.)
- **Nothing is inferred silently.** Unparsed text is reported, ambiguity becomes asks, every reading ships its audit trail (B1 §Analysis 7; C2 P10).

# The fast learning brain

The brief had two halves: learn *fast* from what the listener does — and never let behaviour from a different day, session, or hour bend what is ranked *right now*. Human moods change quickly, so the learning must forget just as quickly. This chapter describes the epoch-v1 layer built for that: epochs, what can move a score within one, what may cross a session boundary, and how isolation is proven. Full mathematics: A3 §3 (P1–P25); code: `apps/brain/src/learning/`.

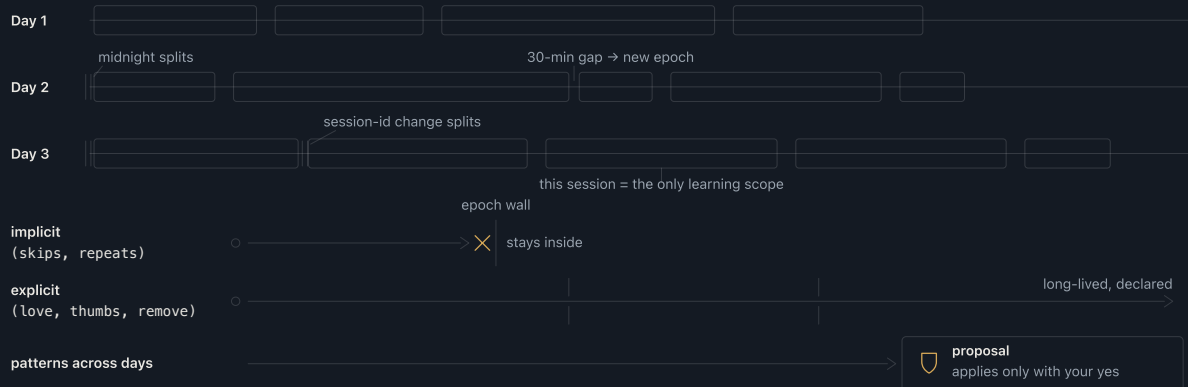


Figure 2 · Three days of listening: what a session is, what stays inside it, and what must ask before crossing.

## Epochs — the unit of "this session"

Learning is only as isolated as its unit of context — and the server's `session_id` is the wrong unit: minted once per installation, it survives restarts and never expires, so it can span days (A4 §2.3). Using it as the session would leak exactly what the requirement forbids.

An **epoch** is derived from the event log instead (A3 §3.1; B2 §3). A boundary exists between two consecutive events when any of three things happens:

- **The session key changes.** Different `session_id`s never mix; id-less events (some other-app captures) form their own bucket rather than joining a session.
- **More than 30 minutes of silence.** A gap of exactly 30:00 keeps the session; 30:01 starts a new one — and since 30 minutes is under an hour, the "or even hour" requirement holds with margin.
- **The local day changes.** Midnight ends an epoch even when only a minute of silence precedes it.

An epoch has a **live window**: for up to 30 minutes after its last event it still counts as "this moment" — resuming within the gap continues the same session; if nothing is live, the interface says so ("no active session — learning is paused"). Each epoch carries a stable id (`ep1:` plus a short hash of its bounds and first event), a daypart label (morning/afternoon/evening/night — context for display and proposals, never a boundary), and its start date.

To the listener: the panel says "This session: night, started 11:30 PM · 14 signals". That is "this session" honestly — the span of continuous listening since the last real break. Restarting the app does not start a new one; walking away for half an hour, or crossing midnight, does.

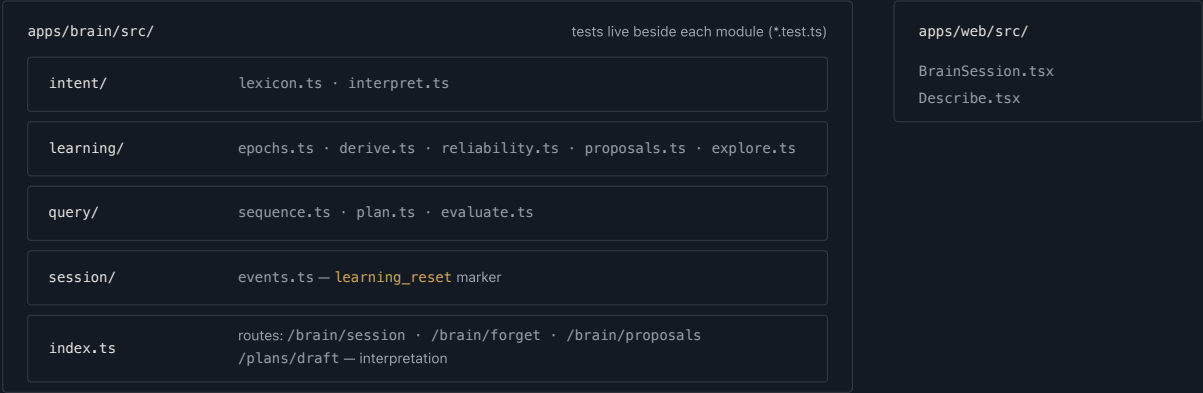


Figure 3 · The new library: intent, learning, sequencing – and where every test lives.

The new code is small and pure: `epochs.ts`, `types.ts`, `derive.ts`, `proposals.ts`, `reliability.ts`, `explore.ts` (see figure).

### What can move a score

All session evidence combines into one value. Each contribution is (magnitude × confidence), decays with a 30-minute half-life inside the session, and is **killed exactly at the epoch boundary** — a mask, not a slower decay (B2 §4). Plain-language evidence table (candidate magnitudes — A3 §3.2):

| WHAT HAPPENED THIS SESSION | PULL         | EXTRA RULE                                                              |
|----------------------------|--------------|-------------------------------------------------------------------------|
| Heard it through           | +0.50 × 0.50 | —                                                                       |
| Replayed it                | +1.00 × 0.70 | —                                                                       |
| Skipped early              | −0.60 × 0.50 | a 2nd early skip of the same track hides it for the rest of the session |
| Skipped late               | −0.10 × 0.20 | weak; never hides                                                       |
| Played in your other apps  | +0.25 × 0.50 | —                                                                       |
| "Not now"                  | −1.00 × 0.90 | hides immediately; nothing persists                                     |

**Hides** are the hard edge of fast forgetting: a repeat-skipped or "not now" track leaves *future* queue builds and resolves while the epoch lives — the current queue is a snapshot that is never reordered. Hides die at the epoch boundary; "Forget this session" clears them immediately — it appends a `learning_reset` marker to the never-edited log, so everything learned in the session stops counting. Deliberate signals — loves, thumbs, removes — survive a reset by design: the sanctioned cross-epoch channel (B2 §5; B5 §2).

Two secondary mechanisms, never dominating. **Artist propagation** lets a track's session evidence touch the artist's other tracks — damped to 35%, capped at ±0.8, opened only by ≥2 events on the artist's other tracks (or a summed magnitude of 1.0); a single event moves almost nothing, and only session-level evidence feeds it (loves do not propagate). It produces the label "other tracks by this session". **Keep/skip centroids** are the one attribute-level learner: from kept tracks and corroborated negatives (repeat-skips, "not now", reason-coded thumbs), it builds small Rocchio-shaped centroids over five produced axes (bpm, loudness, crest factor, onset rate, percussiveness). Gates: ≥3 keeps, ≥2 rejects, ≥60% coverage, spread-based shrinkage;  $\beta = 0.5$  vs  $\gamma = 0.15$  ( $\gamma < \beta$  per the IR-textbook convention); per-axis parts ≤ ±0.5, feature term ≤ ±1.0. Missing measurements contribute nothing and say why — null is never zero (A3 §3.3c; B2 §4).

Composition happens exactly once: **bonus = 0.15 · tanh((persistent + session + artist + features)/2)**. One love reaches ≈0.114 of the 0.15 maximum; twenty loves cannot exceed it (flood-tested — C1 §3). The bonus applies only inside the strict tier: it can re-order and re-select within what a request allows, but cannot break a hard rule (an excluded track stays excluded under maxed-out evidence; C1 §3). The learning meaning changed, so the version is `epoch-v1`; switching back to `legacy-v1` re-derives the same immutable log — never a migration (B5 §1).

## Proposals — the only cross-session channel

Anything outliving a session must pass a human gate. **Proposals** scan closed epochs within 30 days: a pattern needs ≥3 epochs across ≥2 calendar dates ("mood ≠ habit" — same-day repetitions alone never propose); "not now" patterns need ≥2 dayparts; positives are stricter (repeat 4/3; other-app plays 5/3). Nothing applies by itself (A3 §3.4; B2 §3). Accepting is an explicit yes: it appends a normal explicit signal — a global thumb — with `detail.proposal_id` for provenance and no synthesized reason (H2 §F3); artist-subject proposals append no event — the decision is recorded and the surface says to follow the artist (B5 §4).

## Isolation — and how it is proven

The guarantee is byte-equality, not a promise: derive for a day-1 moment, append arbitrary day-2 implicit events, derive again — byte-identical output; day-1 implicit events never move day-2 values (B2 §2; C1's probes: 20/0). The suite also exercises boundaries (29:59 keeps / 30:00 exact keeps / 30:01 splits; midnight with a 60-second gap), flood bounds (20 loves / 20 skips ⇒ |bonus| ≤ 0.15), determinism (reversed and interleaved insertion orders byte-identical; lab re-runs identical, hash 6532f3f3...37ab3a), hide expiry at exactly +30:00, reset semantics, and proposal boundaries (2 epochs ⇒ none; 3/2 ⇒ exactly one). C1 re-ran all of it adversarially from the code (probe batches 20/0, 41/0, 33/0); brain-lab scenarios cover the same ground: **05-fast-learning** (within-epoch stream reorders the next resolve; hides re-skips), **06-isolation** (same stream at now+16h: `learning-adjust-zero`, `learning-hides-empty`, `learning-resolve-identical`), **07-forget** (`learning_reset` clears the epoch's evidence).

## Exploration — built, off by default

The layer ships a deterministic exploration hook, **off by default**; when off, the derived view is byte-identical to a build without exploration (B2 §6). Reason: for a single listener, exploration noise is indistinguishable from a mood change — the exact confusion the isolation work prevents; determinism and honest exposure records are product promises; the novelty/satiation trade-off is one only the user can strike (A3 §3.5). If ever enabled by the user: one slot, seeded from `hash(epoch_id, counter)`, recorded as such.

## Honest limits

- All magnitudes are **candidate values** from the dossier/v1 lineage — hypotheses, not tuned evidence; nothing is tuned on real listening; fixtures are synthetic (A3 §6; B2 §7).
- Reliability weights exist only for **bpm** (`tempo_confidence`; spot-checks write full confidence); other axes use placeholder 1.0.
- Proposal thresholds are conservative, untested candidates; failure mode: too few proposals, never wrong automatism.
- Axis collinearity is contained — small skip weight, corroboration gates, honest notes — not solved.
- Two single-switch calls stay open: strict `global_only` scope persistence, and the epoch-wide skip-hide.

# Bias, culture, and measurement honesty

## 3.1 DOMINANT-CULTURE BIAS IS A CORRECTNESS PROBLEM

Every default in a playlist brain is a claim about how music works. SynAmp inherits defaults built for Western, equal-tempered, 4/4 pop: one global tempo per track, key as a twelve-tone template, loudness as a stand-in for intensity, "energy" as one dial. On a real library they are confidently wrong — and learnable. Bias here is a correctness problem: measurements are a moving target, and a system that will not admit it builds lists on claims it cannot support [A2 §0, §3].

Counterexamples:

- **Ewe timeline patterns.** The gankogui bell carries the twelve-pulse standard pattern <2212221>; 3:2 relations underpin typical West African polyrhythmic textures [CC-1, CC-2]. Several true pulses coexist, so onset analysis can lock onto the wrong layer — low pulse clarity can describe fully rhythmic music. Ambiguity is not "no beat" (CC-1 full text UNRETRIEVED; CC-2 tertiary).
- **Indian tala.** Metre is cyclic and additive — the same 14 beats group differently (Jhoomra 3+4+3+4 vs Dhamar), the sam carries the weight, tempo is elastic [CC-6]. State-of-the-art beat tracking, meter estimation and downbeat detection on Carnatic, Hindustani and Turkish material is "not adequate" [CC-38].
- **Gamelan.** Slendro and pelog are per-ensemble tunings ("island to island, village to village") with deliberate paired-instrument beating; 12-TET chroma models misreport key, and interlocking kotekan "sounds faster than any single human could possibly play" — a double-time trap [CC-8, CC-10, CC-11].
- **Sámi yoik — ontology, not genre.** Yoik is wordless and often unaccompanied — "singing into being" a person, landscape or animal, not singing about one [CC-16, CC-17]. The right behaviour is abstention plus artist-provided descriptions.

The machine side is a moving target. Tempo estimation has a named octave-error class (60/120/240), and its standard metric forgives factors 2, 3,  $\frac{1}{2}$ ,  $\frac{1}{3}$  [CC-37]. Dataset skew is systematic: canonical tempo sets are Western/electronic-heavy with documented mislabels; similarity models diverge from human cross-cultural judgments across nine traditions [CC-42]. SynAmp's own case: 7,351 onsets, phase concentration  $\approx 0.034$  — pulse trackable, timing withheld [INT-2]. "Energy carried by microtiming" stays a labelled candidate [CC-46].

The listener is enculturated too: 12-month-olds respond culture-specifically; adults without exposure miss foreign non-isochronous metres; the groove inverted-U inverts outside 4/4 for Western listeners [CC-29, CC-30, CC-35, CC-36]. Universal "energy" semantics is unsupported at both ends of the chain [A2 §2].

## 3.2 WHAT SHIPPED, IN RESPONSE

The response is honesty machinery, not cleverer scores.

- **No universal "energy".** Energy asks compile to labelled proxies with named inputs (tempo, onsets, percussiveness, loudness) — loudness smallest-weighted, because LUFS is a normalisation standard, not felt intensity [CC-43–45; C2 P3].
- **Soft-only for declared fields.** Mood, arousal, valence, danceability and the other unproduced fields never supply measurements — constraints stay soft, carrying the "no producer yet" note [C2 P10]. Documented caveat: neutral terms still enter the soft average as a constant-0.5 saturation, shifting feedback and MMR margins [H3 E1].
- **Reliability gates.** BPM is gated by tempo confidence (full  $\geq 0.5$ ; damped with a note below; refused at  $\leq 0.2$  or missing); beat-stage fields are timing-gated. "A method's rejection is not independent musical truth": rejected tracks are never called arrhythmic — rejection lowers confidence and asks, not asserts [C2 P9; INT-2, INT-3].



**Figure 4** · Measurement honesty: which fields are trusted, which are gated, and why unknown is never zero.

- **"Tempo unclear — tap it", partially shipped.** The spot-check is live (right / half / double / counted-in-threes / tap-to-replace / "no steady beat"), and user-verified values outrank measured ones — a correction holds while the measured BPM still matches [INT-1]. Tap-the-cycle for 5/7/9/11 is planned [C2 S2].
- **Never auto-4/4.** No metre field is emitted, so nothing asserts 4/4 or any bar length; no bar-level hard rules [C2 P11].
- **No auto genre→culture claims.** Tags are weak hints; "latin party music" reads dance from "party" and asks about "latin"; no "Latin = danceable" claim [C2 P2, E1].
- **Culture-flavoured defaults, labelled.** No tradition-named default ships at all [A2 §4.3]; what ships is the calibration line — "Tempo/energy bands are research starting points, not universal — they stay soft and learn from your session feedback and spot-check corrections." Culture notes are code metadata, unrendered [H3 E3; C2 F7].

### 3.3 ETHICS GUARDRAILS AS DESIGN CONSTRAINTS

The anchors are named: CARE, with its shift "from regulated consultation to value-based relationships" [CC-47]; UNDRIP rights over cultural heritage and traditional cultural expressions, including free, prior and informed consent [CC-49]; and the Indigenous Protocol & AI paper's insistence on no single unified statement [CC-48]. In practice: public sources only; no sacred or restricted knowledge; no ceremonial inference from audio; attribution in the appendix, which names the traditions used here specifically (Ewe; Sámi; gamelan/kotekan; Yolŋu manikay; Māori taonga pūoro; others) [H3 E3]. Nothing crosses a session without an explicit yes; exploration ships off; adjustments are bounded ( $\pm 0.15$ ), explainable, reward-free; mood is never inferred from behaviour — only explicit sad or angry language opens catharsis [C2 P4–P6, S10, S14; H1-8].

### 3.4 BIAS STATEMENTS, QUOTABLE

Twelve lines adapted from C2's inventory; each is true of the shipped slice and quotable as-is.

1. **Bands are starting points, never laws:** "Tempo/energy bands are research starting points, not universal — they stay soft and learn from your session feedback and spot-check corrections." [CC-29/30/36/37]
2. **Tempo is the weakest signal:** octave errors (60/120/240) are a named MIR failure class; low-confidence tempo is damped or refused, never stated as fact; tap-the-cycle remains planned. [CC-37; INT-1; C2 S2]
3. **No universal "energy":** asks compile to labelled proxies with named inputs; loudness smallest-weighted because LUFS is not perceived intensity. [CC-43–45]

4. **No genre→culture claims:** tags are weak hints; "latin party music" reads dance, not culture, and asks about "latin". [C2 P2]
5. **Sensitive words are never product language:** "tribal", "primitive", "exotic", "gypsy", "world music" appear nowhere in shipped code or copy — only echoed back from the user's own prompt. [C2 P1]
6. **Attribution, not absorption — with a caveat:** the appendix names traditions and peoples; shipped code cites rule ids, names no one, and culture-note rendering stays open. [H3 E3; C2 F1]
7. **No invented traditions, no sacred-knowledge inference:** ceremonial function is never inferred from audio; no tradition-named default ships before consultation and a decision record. [A2 §4.2–4.3]
8. **Quiet-time and ceremony is a named gap, not a feature:** no ceremonial-flag mechanism exists; user exclusions stay hard; a user-taggable context flag is the recommended next step. [C2 F10]
9. **Learning is epoch-scoped, not person-scoped; nothing crosses a session without a yes:** implicit behaviour hard-masks at the session boundary; only deliberate signals and accepted proposals cross. [C2 S9, S10]
10. **Suspect measurements can't teach feature tastes:** a skip on low-reliability material is identity-level evidence only, never a tempo or rhythm hypothesis — a method's rejection is not musical truth. [A2 §5 r12; INT-2]
11. **Null is never zero:** unmeasured fields stay unknown with "not measured" reasons; a missing tag is not absence. [C2 S13]
12. **Where a protection isn't built, the report says so:** still open — ceremony flags, metre-correction depth, angry-word coverage (added after review; not user-tested). [C2 S15; H1-8]

Boundary note: behaviour claims rest on code and probes on a synthetic fixture — none is evidence about real music; A2's UNRETRIEVED list stays un-upgraded.

# Verification, rollback, and handoff

Verification here means three re-runnable commands, counts captured fresh on the frozen tree, adversarial review, route smoke against a live server — and notes wherever a check cannot see.

## 4.1 Baseline → final canon

| VERIFICATION                                                          | BASELINE                                | FINAL CANON (FRESH CAPTURE AT WRITE TIME)                                  |
|-----------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------|
| Brain suite ( <code>pnpm --filter @synamp/brain test</code> )         | <b>149</b> — 147 pass · 1 fail · 1 skip | <b>218</b> — 216 pass · 1 fail · 1 skip                                    |
| Type-checks ( <code>@synamp/brain</code> , <code>@synamp/web</code> ) | both exit 0                             | both exit 0                                                                |
| brain-lab harness                                                     | (did not exist)                         | <b>8 scenarios · 86 pass · 0 fail · 0 pending · 5 documented skips</b>     |
| C1 adversarial probes (five scripts)                                  | —                                       | 132 checks; one red = the <code>wave</code> bug (fixed, regression-pinned) |

Both reds are carried, not earned: the fail is a macOS case-fold artifact — the librarian artist-merge expects `Ani DiFranco/...` after renaming `Ani Difranc0/...`, which case-insensitive APFS cannot deliver (passes on Linux CI); the skip is the cross-disk `/dev/shm` case, absent on macOS. Same test, same skip, every run; 69 tests were added net with zero new reds. (The root `type-check` chain still stops in the pre-broken vendored `webamp`.) Counts above are the freshest capture at write time (C4's fresh re-runs, 2026-10-07, post H1–H4); the delivery manifest re-captures at S5 and supersedes them. Intermediates (205 / 64 / 66) are never quoted.

## 4.2 The harness: `tools/brain-lab`

A read-only harness loads the repo's real modules (absent ones reported, never stubbed) and drives eight scenarios with pinned inputs — clock, fixture, ids. One command:

```
cd ~/dev/web-apps/SynAmp
node --experimental-strip-types tools/brain-lab/lab.mts
```

`--json <path>` writes machine evidence ( `synamp.brain-lab/1` ). What they prove:

- **01–04** — flagship phrases yield valid readings (pump/dance run the full evaluate path, strict 25/25); vague/impossible return asks, never dead ends; contradictions give two labelled readings.
- **05 fast-learning** — inside one epoch, repeat-skips hide tracks ( `counts.hidden_by_session = 2` , no dead Restore), kept tracks re-rank (#33→#0, #23→#2, #39→#1); four `sequence-wave-*` regression checks.
- **06 isolation** — the same stream at `now+16h` : adjustments `+0.000` , hides empty, resolve **byte-identical** to baseline. Proven, not promised.
- **07 forget** — the `learning_reset` marker clears the epoch's evidence.
- **08 culture-guardrails** — banned product words absent (user echo counted), culture spans never asserted outside quoted echo, no dead ends — a pin over honest behaviour, not classifier accuracy.

**Determinism and CWD.** Two runs are byte-identical modulo `generated_at` ; a run from `/tmp` normalizes to the same hash ( `c81b8753...` ). When replayability is the promise, a harness drifting with clock or directory tests the environment, not the code; C4's fresh re-run matched all four canonical captures.

**The five skips** are registered, not silent: two mark the legacy pipeline legs (coverage moved to `intent-chosen-evaluates`); three are not-applicable by design — evaluating a vague or impossible ask is a category error; the contract is asserted instead.

**Honest boundary.** The lab composes modules exactly as `index.ts` `policyOptions()` does but cannot start the server; routes are covered below. Fixtures are synthetic — no evidence about real music.

### 4.3 Route-level evidence

**Auth on/off.** `/health` public; without a token the brain routes, `/events` and `/settings` return 401, with it 200 (bad bodies still 400); `/api/v1/brain` is protected.

**Policy rollback, live.** One POST — `{"listening_policy":"legacy-v1"}` — flips the readout to `policy_version: heuristic-v1`, `epoch: null`, `hides: []`; the queue adjustment falls from `-0.5999` ("skipped early this session") to 0. Switching back restores the hide — a pure re-derive from the untouched log. No restart, no migration (C4 live).

**Forget.** POST `/brain/forget` appends a `learning_reset` marker — append-only proven: 14→15 log lines, first-line hash unchanged.

**Error paths** (probed live; no 500s, nothing leaked):

| PROBE                                                    | RESULT                                             |
|----------------------------------------------------------|----------------------------------------------------|
| forget <code>scope:"all"</code> / <code>"session"</code> | 400 — "the event log is never edited, only marked" |
| forget empty body / invalid JSON                         | 400                                                |
| proposals missing id / bad action                        | 400                                                |
| proposals unknown id                                     | 404 — "No suggestion with that id"                 |
| draft prompt >500 chars / whitespace-only                | 400 (500 chars boundary → 200)                     |
| settings bogus policy                                    | 400                                                |

**Queue/resolve smokes.** "pump me up..." resolves with an identical order at `resolve == explain == evaluate`, `sequencing_applied` at all three; a skip hides `sample-034` and the list re-derives (same-artist `sample-033` drops too — deterministic); draft previews match; screenshots from the built dist.

### 4.4 What review caught — and fixed

C1's probes found the sharpest catch — `wave` corrupting odd measured counts (`n=3` → `[hi, lo, undefined]` → `null` → queue-build 500); it survived a green unit suite because the wave test used even tracks and the determinism loop compared outputs to themselves. A coverage hole, closed. H1 fixed it, added `n=1/3/5` regression tests (even counts byte-identical); H4 converted it into lab checks. Display/copy: session hides no longer masquerade as "Removed by you" (no dead Restore); accepted proposals no longer synthesize a reason (`detail.proposal_id` provenance); consequences precede the click; "% sure" → confidence words; sequencing notes render. Docs: declared-field caveat, propagation marker, guardrails attribution, D8, counts hygiene. Counts were re-captured, not re-quoted.

### 4.5 Rollback and operations

C4's verdicts: **rollback — YES; error paths — YES; delivery inputs — YES**, with the commit precondition as its only High finding.

- **Branch level.** The slice ships as a single commit on `feat/fast-learning-brain` ; revert = `git reset --hard <base>` (or `git checkout main` ) — no migrations; data files are additive and inert under an old build; hashes go in the manifest. *Precondition (C4 F1, closed at S5):* the tree was uncommitted at review time and the patch command void; S5 commits first.
- **Policy switch.** `POST /api/v1/settings {"listening_policy":"legacy-v1"}` and back — instant; curl-only (no UI — carried).
- **Forget.** `POST /api/v1/brain/forget` — append-only; loves/thumbs/removes survive by design.
- **Proposals store.** `brain-proposals.json` (atomic, 0600); decisions never expire — a dismissed id stays suppressed (documented limitation).
- **Exploration stays off** (byte-identical when off).
- **Scratch recipe.** `PLAYLIST_DATA_PATH=... LIBRARY_SIGNALS_PATH=... BRAIN_PORT=... node --experimental-strip-types src/index.ts` — state derives beside the playlist store; no `BRAIN_DATA_DIR` ; no new envs; no telemetry.
- **Environment notes.** Dev-sandbox residue (`apps/brain/data/` , 14 synthetic events, gitignored) stays, documented; stale user servers on :5173:3001 are not ours to touch.

## 4.6 Next smallest experiments — and a status ledger

1. **One real-library session, then a tuning pass.** Read `GET /api/v1/brain/session` after real listening; magnitudes/thresholds are labelled candidates — isolation is proven, "does it feel right" is not.
2. **P4 producers.** `vocal_fraction` , `instruments.*` , mood/arousal/valence/danceability are declared but unproduced; lexicon terms stay inert until producers land.
3. **Proposal thresholds** — conservative, untested guesses (3 epochs / 2 dates); revisit after real data.
4. **Metre-correction UI depth** — the tap path seeds the human-verified tier; a surfaced UI (5/7/9/11 tap-the-cycle), like ceremony flags, remains open.

**Status ledger.** *Confirmed:* counts, determinism, rollback levers, error paths, auth, no telemetry.

*Partial:* proposal accept — the 409 path is not freshly re-exercised (B5/H2 receipts cover it); the legacy policy path is route-verified, not lab-modelled. *Unverified:* real-music behaviour and long-run logs.

Handoff: the `DELIVERY/verification/` manifest re-captures counts and checksums on the frozen tree; `review.md` carries the round conclusions; the branch — one commit — is the live copy.

# Appendix — sources, parameters, glossary

*Sources were retrieved by the A-round researchers on 2026-10-06; this appendix only re-organises their records — it adds nothing and fetches nothing.*

**Contents** - §5.1 Sources appendix (merged MP / CC / ML, original IDs preserved) — incl. §5.1.5 UNRETRIEVED / blocked - §5.2 Parameter table P1–P25 — annotated with implementation + review outcomes - §5.3 Glossary (16 terms, plain language) - §5.4 Reproduction commands (full verification stack)

## 5.1 Sources appendix — merged [MP-n] / [CC-n] / [ML-n] (original IDs preserved)

Reading notes (apply to all three tables):

- **IDs:** [MP-1]–[MP-34] = music psychology (A1 §6); [CC-1]–[CC-52] = cross-cultural & metric reliability (A2 §6); [ML-1]–[ML-23] = learning science / recommender ML (A3 §6). Do not renumber — other chapters cite these IDs.
- **Retrieval mode** is reproduced as the packs stated it: full text / abstract / via API (Semantic Scholar, OpenAlex, Europe PMC, PubMed E-utilities) / snippet / tertiary (Wikipedia, Britannica, official docs). A1 gave locators (PMC/PubMed IDs, "full", "abstract") rather than URLs where noted; **no URLs were invented** for this merge.
- **Tiers** are as assigned by each pack: High = ≥2 independent retrieved sources agreeing; Medium = one authoritative retrieved source; Low = weak/tertiary single source; Conflict = sources disagree. A1 tiers at claim-cluster level, so MP rows show the cluster label for the claim the source supports ("—" = no individual tier stated by A1).
- **Dedup check (this merge):** no exact duplicates across the three families (URL/DOI/arXiv cross-check); each ID maps to one entry. [CC-35b] is a companion press summary inside the [CC-35] row.
- **Conflicts are kept, not smoothed** — rows involved in a listed conflict carry a "Conflict #n" note (A1 §6 conflict list) or the A2 caution verbatim (e.g., "same publisher family — treat as one").

### 5.1.1 [MP-N] — MUSIC PSYCHOLOGY (A1 §6; RETRIEVALS 2026-10-06)

| ID   | SOURCE (AUTHORS, YEAR — TITLE — VENUE)                                                                                                                                                     | RETRIEVED AS               | TIER                              | CLAIM CLUSTER IT SUPPORTS                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MP-1 | Karageorghis & Priest (2012) — Music in the exercise domain: a review and synthesis (Part II) — Int. Review of Sport and Exercise Psychology 5(1), 67–84                                   | PMC3339577 (full)          | High (exercise cluster)           | Pre-task arousal/imagery; async band 125–140 bpm (40–90 % max HRR); stimulative >120 bpm percussive; sedative <80 bpm, 60–70 guidance; synchronous benefits. (Part I UNRETRIEVED.) |
| MP-2 | Terry, Karageorghis, Curran, Martin & Parsons-Smith (2020) — Effects of music in exercise and sport: a meta-analytic review — Psychological Bulletin 146(2), 91–117                        | BURA green-OA PDF (full)   | High (exercise)                   | Affect g=0.48; performance g=0.31; RPE g=0.22; VO <sub>2</sub> g=0.15; HR ns; tempo moderation.                                                                                    |
| MP-3 | Delleli, Ouergui, Ballmann, Messaoudi, Trabelsi, Ardigò & Chtourou (2023) — Pre-task music effects... systematic review with multilevel meta-analysis — Frontiers in Psychology 14:1293783 | PMC10701429 (full)         | High (exercise)                   | Pre-task gains (time, power, fatigue, feeling); self-selected > pre-selected; wide prediction intervals.                                                                           |
| MP-4 | Juslin & Västfjäll (2008) — Emotional responses to music: the need to consider underlying mechanisms — Behavioral and Brain Sciences 31, 559–575                                           | PubMed 18826699 (abstract) | Covered (abstract; no tier given) | Six-mechanism framework; emotions via non-music-specific mechanisms.                                                                                                               |

| ID    | SOURCE (AUTHORS, YEAR — TITLE — VENUE)                                                                                                                                                         | RETRIEVED AS                                                         | TIER                       | CLAIM CLUSTER IT SUPPORTS                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------|
| MP-5  | Juslin (2013) — From everyday emotions to aesthetic emotions — Physics of Life Reviews 10(3), 235–266                                                                                          | PubMed 23769678 (abstract)                                           | Covered (abstract)         | BRECVEMA mechanisms; musical event = music × listener × context.                                         |
| MP-6  | communicationtheory.org — Mood Management Theory (Zillmann)                                                                                                                                    | secondary page                                                       | Low (tertiary secondary)   | Media chosen to optimise mood; <b>Zillmann 1988 primary UNRETRIEVED.</b>                                 |
| MP-7  | Heiderscheit & Madson (2015) — Use of the iso principle as a central method in mood management — Music Therapy Perspectives 33(1), 45–52                                                       | <b>UNRETRIEVED</b> (metadata verified via search + Semantic Scholar) | —                          | Covered by [MP-8][MP-9].                                                                                 |
| MP-8  | Starcke et al. (2021) — Emotion modulation through music after sadness induction — Int J Environ Res Public Health 18(23), 12486                                                               | PMC8656869 (full)                                                    | Medium                     | Sad→happy sequence: highest positive / lowest negative affect; not all contrasts significant.            |
| MP-9  | Cheung et al. (2025) — Individualized iso-principle playlist for de-escalating agitation (RCT feasibility) — Int J Geriatr Psychiatry 40(4), e70070                                            | PMC11949771 (full)                                                   | Conflict #4                | Feasible; <b>not more effective than control</b> ; efficacy unconfirmed.                                 |
| MP-10 | Witek, Clarke, Wallentin, Kringelbach & Vuust (2014) — Syncopation, body-movement and pleasure in groove music — PLOS ONE 9(4), e94446                                                         | full                                                                 | Medium; Conflict #7        | Inverted-U; medium syncopation max wanting/pleasure; dancers stronger; entropy poor.                     |
| MP-11 | Janata, Tomic & Haberman (2012) — Sensorimotor coupling in music and the psychology of the groove — J Exp Psychol Gen 141(1), 54–75                                                            | <b>UNRETRIEVED</b> (closed; metadata via Semantic Scholar)           | —                          | Groove foundational text.                                                                                |
| MP-12 | Stupacher, Hove & Janata (2016) — Audio features underlying perceived groove and sensorimotor synchronization — Music Perception 33(5), 571–589                                                | <b>UNRETRIEVED</b> (closed; record via search)                       | —                          | Groove features.                                                                                         |
| MP-13 | Hine, Wakana & Nakauchi (2025) — How we remember music tempo: spontaneous motor tempo in recall and preference — Frontiers in Psychology 16:1631625                                            | PMC12549237 (full)                                                   | Medium                     | SMT predicts adjusted preferred tempo.                                                                   |
| MP-14 | Stupacher, Witek, Vuoskoski & Vuust (2020) — Cultural familiarity and individual musical taste... social bonding when moving to music — Scientific Reports 10 (doi 10.1038/s41598-020-66529-1) | PMC7308378 (full)                                                    | Medium                     | Synchrony × enjoyment drives closeness; familiarity vs enjoyment dissociate.                             |
| MP-15 | Sloboda (1991) — Music structure and emotional response: some empirical findings — Psychology of Music 19(2), 110–120                                                                          | <b>UNRETRIEVED</b> (paywall; mirrors blocked)                        | —                          | Frisson/chills classic.                                                                                  |
| MP-16 | Blood & Zatorre (2001) — Intensely pleasurable responses to music correlate with activity in brain regions implicated in reward and emotion — PNAS 98(20), 11818–11823                         | PMC58814 (full)                                                      | High (frisson)             | Chills + autonomic changes; ventral striatum, midbrain, amygdala, OFC, VMPFC.                            |
| MP-17 | Sachs, Damasio & Habibi (2015) — The pleasures of sad music: a systematic review — Frontiers in Human Neuroscience 9:404                                                                       | full                                                                 | Conflict #2                | Sadness pleasurable when non-threatening/aesthetic/benefit-producing.                                    |
| MP-18 | Taruffi & Koelsch (2014) — The paradox of music-evoked sadness: an online survey — PLOS ONE 9(10), e110490                                                                                     | full                                                                 | Conflict #2                | 4 rewards; mood-congruent; nostalgia most frequent.                                                      |
| MP-19 | Garrido & Schubert (2015) — Moody melodies: do they cheer us up? — Psychology of Music 43(2), 244–261                                                                                          | SAGE abstract page                                                   | Conflict #2                | Increased depression after self-selected sad music.                                                      |
| MP-20 | Perham & Currie (2014) — Does listening to preferred music improve reading                                                                                                                     | full PDF (mirror)                                                    | High (lyrics); Conflict #1 | Liked = disliked lyrical music, both worse than non-lyrical/quiet; ISE needs changing-state + seriation. |

| ID                                                                                | SOURCE (AUTHORS, YEAR — TITLE — VENUE)                                                                                                                                                       | RETRIEVED AS                               | TIER                                       | CLAIM CLUSTER IT SUPPORTS                                                                                                                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   | comprehension performance? — Applied Cognitive Psychology 28(2), 279–284                                                                                                                     |                                            |                                            |                                                                                                                                          |
| MP-21                                                                             | Souza & Barbosa (2023) — Should we turn off the music? Music with lyrics interferes with cognitive tasks — Journal of Cognition 6(1), 24                                                     | PMC10162369 (full)                         | High (lyrics)                              | Lyrics d $\approx$ -0.3 verbal/visual memory + reading; arithmetic ns; instrumental neutral.                                             |
| MP-22                                                                             | Kämpfe, Sedlmeier & Renkewitz (2011) — The impact of background music on adult listeners: a meta-analysis — Psychology of Music 39(4), 424–448                                               | ERIC EJ944201 (abstract)                   | High (lyrics) / Weak (chores); Conflict #1 | Global null; reading disturbed; small memory detriments; positive emotional reactions; tempo paces activity.                             |
| MP-23                                                                             | Husain, Thompson & Schellenberg (2002) — Effects of musical tempo and mode on arousal, mood, and spatial abilities — Music Perception 20(2), 151–171                                         | <b>UNRETRIEVED</b> (SAGE blocked)          | —                                          | Tempo→arousal, mode→mood (publisher record; framework in [MP-32]).                                                                       |
| MP-24                                                                             | Pietschnig, Voracek & Formann (2010) — Mozart effect–Shmozart effect: a meta-analysis — Intelligence 38(3), 314–323                                                                          | <b>UNRETRIEVED</b> (ScienceDirect captcha) | —                                          | Covered by [MP-25].                                                                                                                      |
| MP-25                                                                             | Oberleiter & Pietschnig (2023) — Unfounded authority, underpowered studies... Mozart effect myth — Scientific Reports 13:3175                                                                | full                                       | Conflict #6                                | No Mozart-specific benefit; publication bias; underpowered studies.                                                                      |
| MP-26                                                                             | Cordi, Ackermann & Rasch (2019) — Effects of relaxing music on healthy sleep — Scientific Reports 9:9079                                                                                     | full                                       | Medium                                     | Subjective sleep 3.69 vs 3.28 (p=.048); N1 reduced; small sample; suggestibility interaction.                                            |
| MP-27                                                                             | Thoma, La Marca, Brönnimann, Finkel, Ehler & Nater (2013) — The effect of music on the human stress response — PLOS ONE 8(8), e70156                                                         | full                                       | Conflict #5                                | Faster sAA recovery; cortisol response highest in music (against hypothesis); HR/subjective ns.                                          |
| MP-28                                                                             | Sedikides, Leunissen & Wildschut (2022) — The psychological benefits of music-evoked nostalgia — Psychology of Music 50(6), 2044–2062                                                        | author PDF (full)                          | High (MEAM existence) / Medium (nostalgia) | Connectedness, self-esteem, optimism, meaning; buffers sadness; familiarity necessity open.                                              |
| MP-29                                                                             | Janata (2009) — The neural architecture of music-evoked autobiographical memories — Cerebral Cortex 19(11), 2579–2594                                                                        | PMC2758676 (full)                          | High + <b>correction</b>                   | MPFC network for MEAMs; subjects selected for $\geq 30$ % MEAM frequency — the "30 %" headline is a selection threshold, not prevalence. |
| MP-30                                                                             | Madison & Schiöde (2017) — Repeated listening increases the liking for music regardless of its complexity — Frontiers in Neuroscience 11:147                                                 | PMC5374342 (full)                          | Conflict #3                                | Monotonic liking increase with repetition (no inverted-U); familiarity strongest predictor.                                              |
| MP-31                                                                             | Li, Chen & Zhang (2019) — Effect of music tempo on long-distance driving — i-Perception 10(4)                                                                                                | full                                       | Medium                                     | Medium tempo best for fatigue/attention; slow worst; single-piece caveat.                                                                |
| MP-32                                                                             | Gigliotti et al. (2025) — The sonic energy of background music impacts cognitive performances — Cognitive Research: Principles and Implications (doi 10.1186/s41235-025-00676-9)             | PMC12627320 (full)                         | — (framework)                              | Both low/high-arousing raise activation; high-arousing raises felt effort; verbal fluency unaffected.                                    |
| MP-33                                                                             | Mas-Herrero, Zatorre, Rodriguez-Fornells & Marco-Pallarés (2014) — Dissociation between musical and monetary reward responses in specific musical anhedonia — Current Biology 24(6), 699–704 | abstract via Europe PMC                    | Medium–High                                | Subgroup with reduced musical pleasure, normal monetary reward (~3–5 % per secondary sources).                                           |
| MP-34                                                                             | Juslin, Liljeström, Västfjäll, Barradas & Silva (2008) — An experience sampling study of emotional reactions to music — Emotion 8(5), 668–683                                                | abstract via Europe PMC                    | —                                          | Music in 37 % of episodes; 64 % of music episodes affected feelings.                                                                     |
| 5.1.2 [CC-N] — CROSS-CULTURAL & METRIC RELIABILITY (A2 §6; RETRIEVALS 2026-10-06) |                                                                                                                                                                                              |                                            |                                            |                                                                                                                                          |

| ID    | SOURCE<br>(AUTHOR/VENUE<br>AS GIVEN)                                                                                                          | RETRIEVED AS                       | URL (AS GIVEN)                                                                                                                                                                                      | TYPE              |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| CC-1  | Agawu —<br>Structural<br>Analysis or<br>Cultural<br>Analysis? ...<br>"Standard<br>Pattern" of West<br>African Rhythm<br>— JAMS 59(1),<br>2006 | Abstract via<br>OpenAlex API       | <a href="https://doi.org/10.1525/jams.2006.59.1.1">https://doi.org/10.1525/jams.2006.59.1.1</a>                                                                                                     | Music<br>(a)      |
| CC-2  | "Ewe music" —<br>Wikipedia                                                                                                                    | Full text                          | <a href="https://en.wikipedia.org/wiki/Ewe_music">https://en.wikipedia.org/wiki/Ewe_music</a>                                                                                                       | Local<br>(t)      |
| CC-3  | "Pygmy music" —<br>Wikipedia                                                                                                                  | Full text                          | <a href="https://en.wikipedia.org/wiki/Pygmy_music">https://en.wikipedia.org/wiki/Pygmy_music</a>                                                                                                   | Local<br>(t)      |
| CC-4  | UNESCO ICH —<br>Polyphonic<br>singing of the<br>Aka Pygmies<br>(no. 00082)                                                                    | Full text                          | <a href="https://ich.unesco.org/en/RL/polyphonic-singing-of-the-aka-pygmyes-of-central-africa-00082">https://ich.unesco.org/en/RL/polyphonic-singing-of-the-aka-pygmyes-of-central-africa-00082</a> | Music<br>(c)      |
| CC-5  | Fürniss — Aka<br>Polyphony,<br>2006 (chapter)                                                                                                 | Abstract via<br>OpenAlex           | <a href="https://doi.org/10.1093/acprof:oso/9780195177893.003.0006">https://doi.org/10.1093/acprof:oso/9780195177893.003.0006</a>                                                                   | Music<br>(a)      |
| CC-6  | "Tala (music)" —<br>Wikipedia                                                                                                                 | Full text                          | <a href="https://en.wikipedia.org/wiki/Tala_(music)">https://en.wikipedia.org/wiki/Tala_(music)</a>                                                                                                 | Local<br>Music    |
| CC-7  | Saraga README<br>(MTG)                                                                                                                        | Fetched<br>(content not<br>quoted) | <a href="https://raw.githubusercontent.com/MTG/saraga/master/README.md">https://raw.githubusercontent.com/MTG/saraga/master/README.md</a>                                                           | —                 |
| CC-8  | "Kotekan" —<br>Wikipedia                                                                                                                      | Full text                          | <a href="https://en.wikipedia.org/wiki/Kotekan">https://en.wikipedia.org/wiki/Kotekan</a>                                                                                                           | Local             |
| CC-9  | "Colotomy" —<br>Wikipedia                                                                                                                     | Full text                          | <a href="https://en.wikipedia.org/wiki/Colotomy">https://en.wikipedia.org/wiki/Colotomy</a>                                                                                                         | Local             |
| CC-10 | "Slendro" —<br>Wikipedia                                                                                                                      | Full text                          | <a href="https://en.wikipedia.org/wiki/Slendro">https://en.wikipedia.org/wiki/Slendro</a>                                                                                                           | Local             |
| CC-11 | "Pelog" —<br>Wikipedia                                                                                                                        | Full text                          | <a href="https://en.wikipedia.org/wiki/Pelog">https://en.wikipedia.org/wiki/Pelog</a>                                                                                                               | Local             |
| CC-12 | "Gamelan" —<br>Wikipedia                                                                                                                      | Full text                          | <a href="https://en.wikipedia.org/wiki/Gamelan">https://en.wikipedia.org/wiki/Gamelan</a>                                                                                                           | Local             |
| CC-13 | "Manikay" —<br>Wikipedia                                                                                                                      | Full text                          | <a href="https://en.wikipedia.org/wiki/Manikay">https://en.wikipedia.org/wiki/Manikay</a>                                                                                                           | Local             |
| CC-14 | "Didgeridoo" —<br>Wikipedia                                                                                                                   | Full text                          | <a href="https://en.wikipedia.org/wiki/Didgeridoo">https://en.wikipedia.org/wiki/Didgeridoo</a>                                                                                                     | Local             |
| CC-15 | "Clapstick" —<br>Wikipedia                                                                                                                    | Full text                          | <a href="https://en.wikipedia.org/wiki/Clapstick">https://en.wikipedia.org/wiki/Clapstick</a>                                                                                                       | Local             |
| CC-16 | Madsen —<br>Singing into<br>Being — UAF<br>Circumpolar<br>Music Series                                                                        | Full text                          | <a href="https://www.uaf.edu/music/cms/singing-into-being.php">https://www.uaf.edu/music/cms/singing-into-being.php</a>                                                                             | Music             |
| CC-17 | Hämäläinen et<br>al. — Sami yoik,<br>Sami history,                                                                                            | Full text<br>(PMC5912196)          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC5912196/">https://pmc.ncbi.nlm.nih.gov/articles/PMC5912196/</a>                                                                                   | Music<br>(p<br>re |

| ID    | SOURCE<br>(AUTHOR/VENUE<br>AS GIVEN)                                                                                    | RETRIEVED AS                                                                     | URL (AS GIVEN)                                                                                                                                                                                                                                        | TYPE              |
|-------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|       | Sami health:<br>narrative review<br>— Int J<br>Circumpolar<br>Health                                                    |                                                                                  |                                                                                                                                                                                                                                                       |                   |
| CC-18 | Te Ara (Flintoff)<br>— Māori musical<br>instruments –<br>taonga puoro                                                   | Full text                                                                        | <a href="https://teara.govt.nz/en/maori-musical-instruments-taonga-puoro">https://teara.govt.nz/en/maori-musical-instruments-taonga-puoro</a>                                                                                                         | M                 |
| CC-19 | Stanfield —<br>Powwow Music<br>and its<br>Polymetric<br>Construction —<br>MUSICultures<br>37, 2010                      | Full PDF                                                                         | <a href="https://journals.lib.unb.ca/index.php/MC/article/view/20228">https://journals.lib.unb.ca/index.php/MC/article/view/20228</a>                                                                                                                 | M<br>H<br>(s<br>p |
| CC-20 | Hoefnagels —<br>Northern Style<br>Powwow<br>Music... —<br>MUSICultures,<br>2004                                         | Abstract                                                                         | <a href="https://journals.lib.unb.ca/index.php/MC/article/view/21605">https://journals.lib.unb.ca/index.php/MC/article/view/21605</a>                                                                                                                 | L                 |
| CC-21 | "Clave (rhythm)"<br>— Wikipedia                                                                                         | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Clave_(rhythm)">https://en.wikipedia.org/wiki/Clave_(rhythm)</a>                                                                                                                                               | L                 |
| CC-22 | "Tresillo<br>(rhythm)" —<br>Wikipedia                                                                                   | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Tresillo_(rhythm)">https://en.wikipedia.org/wiki/Tresillo_(rhythm)</a>                                                                                                                                         | L                 |
| CC-23 | "Aksak (meter)"<br>— Wikipedia                                                                                          | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Aksak">https://en.wikipedia.org/wiki/Aksak</a>                                                                                                                                                                 | L<br>M            |
| CC-24 | Britannica —<br>aksak                                                                                                   | Full text                                                                        | <a href="https://www.britannica.com/art/aksak">https://www.britannica.com/art/aksak</a>                                                                                                                                                               | M                 |
| CC-25 | "Huayno" —<br>Wikipedia                                                                                                 | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Huayno">https://en.wikipedia.org/wiki/Huayno</a>                                                                                                                                                               | L                 |
| CC-26 | "Siku<br>(instrument)" —<br>Wikipedia                                                                                   | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Siku_(instrument)">https://en.wikipedia.org/wiki/Siku_(instrument)</a>                                                                                                                                         | L                 |
| CC-27 | "Ma (negative<br>space)" —<br>Wikipedia                                                                                 | Full text                                                                        | <a href="https://en.wikipedia.org/wiki/Ma_(negative_space)">https://en.wikipedia.org/wiki/Ma_(negative_space)</a>                                                                                                                                     | L                 |
| CC-28 | Britannica — jo-<br>ha-kyū /<br>Japanese music<br>structural ideals                                                     | Topic stub                                                                       | <a href="https://www.britannica.com/art/jo-ha-kyu">https://www.britannica.com/art/jo-ha-kyu</a>                                                                                                                                                       | L<br>M            |
| CC-29 | Hannon &<br>Trehub — Tuning<br>in to musical<br>rhythms: infants<br>learn more<br>readily than<br>adults — PNAS<br>2005 | Full text<br>(PMC1194930)                                                        | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC1194930/">https://pmc.ncbi.nlm.nih.gov/articles/PMC1194930/</a>                                                                                                                                     | M<br>(p<br>re     |
| CC-30 | Hannon &<br>Trehub —<br>Metrical<br>Categories in<br>Infancy and<br>Adulthood                                           | Full PDF<br>(CogSci 2003)<br>+ Psych<br>Science 2005<br>abstract via<br>OpenAlex | <a href="https://escholarship.org/content/qt6170j46c/qt6170j46c.pdf">https://escholarship.org/content/qt6170j46c/qt6170j46c.pdf</a> + <a href="https://doi.org/10.1111/j.0956-7976.2005.00779.x">https://doi.org/10.1111/j.0956-7976.2005.00779.x</a> | M                 |

| ID    | SOURCE<br>(AUTHOR/VENUE<br>AS GIVEN)                                                                                             | RETRIEVED AS                                    | URL (AS GIVEN)                                                                                                                                                                                                                                                                                                                                                                                          | TYPE           |
|-------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| CC-31 | Jacoby & McDermott — Integer Ratio Priors on Musical Rhythm — Current Biology 2017                                               | Abstract via PubMed E-utilities (PMID 28065607) | <a href="https://pubmed.ncbi.nlm.nih.gov/28065607">https://pubmed.ncbi.nlm.nih.gov/28065607</a>                                                                                                                                                                                                                                                                                                         | Music (a)      |
| CC-32 | Jacoby et al. — Cross-Cultural Work in Music Cognition — Music Perception 37(3), 2020                                            | Full PDF                                        | <a href="https://www.norijacoby.com/Jacoby_etal_20202_cross%20cultural%20work%20in%20music%20cognition.pdf">https://www.norijacoby.com/Jacoby_etal_20202_cross%20cultural%20work%20in%20music%20cognition.pdf</a>                                                                                                                                                                                       | Music (p<br>pa |
| CC-33 | Mehr et al. — Universality and diversity in human song — Science 2019                                                            | Full text (PMC7001657)                          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC7001657/">https://pmc.ncbi.nlm.nih.gov/articles/PMC7001657/</a>                                                                                                                                                                                                                                                                                       | Music (p<br>re |
| CC-34 | Savage et al. — Statistical universals... — PNAS 2015                                                                            | Full text (PMC4517223)                          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC4517223/">https://pmc.ncbi.nlm.nih.gov/articles/PMC4517223/</a>                                                                                                                                                                                                                                                                                       | Music (p<br>re |
| CC-35 | Polak et al. — Rhythmic Prototypes Across Cultures — Music Perception 2018 <b>(incl. companion [CC-35b] MPI press summary)</b>   | Abstract via OpenAlex + press release           | <a href="https://doi.org/10.1525/mp.2018.36.1.1">https://doi.org/10.1525/mp.2018.36.1.1</a> ; <a href="https://www.aesthetics.mpg.de/en/newsroom/news/news-article/article/rhythm-diversity-different-cultures-prefer-different-beat-patterns-1.html">https://www.aesthetics.mpg.de/en/newsroom/news/news-article/article/rhythm-diversity-different-cultures-prefer-different-beat-patterns-1.html</a> | Music (a<br>re |
| CC-36 | Spiech et al. — "4/4 and more..." rhythmic complexity predicts groove in common meters — Communications Psychology (Nature) 2025 | Full PDF                                        | <a href="https://www.nature.com/articles/s44271-025-00360-0.pdf">https://www.nature.com/articles/s44271-025-00360-0.pdf</a>                                                                                                                                                                                                                                                                             | Music (p<br>re |
| CC-37 | Schreiber & Müller — Music Tempo Estimation: Are We Done Yet? — TISMIR 2020                                                      | Full text                                       | <a href="https://transactions.ismir.net/articles/10.5334/tismir.43">https://transactions.ismir.net/articles/10.5334/tismir.43</a>                                                                                                                                                                                                                                                                       | Music (p<br>re |
| CC-38 | Srinivasamurthy et al. — In Search of Automatic Rhythm Analysis Methods for Turkish and Indian Art Music                         | Abstract via OpenAlex                           | <a href="https://doi.org/10.1080/09298215.2013.879902">https://doi.org/10.1080/09298215.2013.879902</a>                                                                                                                                                                                                                                                                                                 | Music (a       |
| CC-39 | Miguel et al. — From beat tracking to beat expectation... — PLoS ONE 2020                                                        | Full text (PMC7673539)                          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC7673539/">https://pmc.ncbi.nlm.nih.gov/articles/PMC7673539/</a>                                                                                                                                                                                                                                                                                       | Music (p<br>re |

| ID    | SOURCE<br>(AUTHOR/VENUE<br>AS GIVEN)                                                                           | RETRIEVED AS           | URL (AS GIVEN)                                                                                                                                                                                                                      | TITLE                              |
|-------|----------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| CC-40 | TISMIR — A Quantitative Model of Tension and Resolution in Turkish Makam Music                                 | Full text              | <a href="https://transactions.ismir.net/articles/10.5334/tismir.353">https://transactions.ismir.net/articles/10.5334/tismir.353</a>                                                                                                 | Musical Tension (p. 353)           |
| CC-41 | Serra et al. — AI and Music at a Crossroads... — TISMIR                                                        | Full text              | <a href="https://transactions.ismir.net/articles/10.5334/tismir.372">https://transactions.ismir.net/articles/10.5334/tismir.372</a>                                                                                                 | Musical Tension (p. 372)           |
| CC-42 | Papaioannou, Benetos & Potamianos — Cross-Cultural Music Similarity... — TISMIR                                | Full text              | <a href="https://transactions.ismir.net/articles/10.5334/tismir.341">https://transactions.ismir.net/articles/10.5334/tismir.341</a>                                                                                                 | Musical Tension (p. 341)           |
| CC-43 | "Loudness" — Wikipedia                                                                                         | Full text              | <a href="https://en.wikipedia.org/wiki/Loudness">https://en.wikipedia.org/wiki/Loudness</a>                                                                                                                                         | Loudness (technical sound concept) |
| CC-44 | "Equal-loudness contour" — Wikipedia                                                                           | Full text              | <a href="https://en.wikipedia.org/wiki/Equal-loudness_contour">https://en.wikipedia.org/wiki/Equal-loudness_contour</a>                                                                                                             | Loudness (technical sound concept) |
| CC-45 | "LKFS" (LUFS) — Wikipedia                                                                                      | Full text              | <a href="https://en.wikipedia.org/wiki/LKFS">https://en.wikipedia.org/wiki/LKFS</a>                                                                                                                                                 | Loudness (technical sound concept) |
| CC-46 | Senn et al. — Expert Performance Microtiming on Listeners' Experience of Groove — Frontiers in Psychology 2016 | Full PDF               | <a href="https://www.frontiersin.org/articles/10.3389/fpsyg.2016.01487/pdf">https://www.frontiersin.org/articles/10.3389/fpsyg.2016.01487/pdf</a>                                                                                   | Musical Tension (p. 1487)          |
| CC-47 | Carroll et al. — The CARE Principles for Indigenous Data Governance — Data Science Journal 2020                | Full text              | <a href="https://datascience.codata.org/articles/10.5334/dsj-2020-043/">https://datascience.codata.org/articles/10.5334/dsj-2020-043/</a>                                                                                           | Historical Research                |
| CC-48 | Lewis et al. (eds.) — Indigenous Protocol and Artificial Intelligence Position Paper, 2020                     | Landing + preview text | <a href="https://www.indigenous-ai.net/position-paper/">https://www.indigenous-ai.net/position-paper/</a>                                                                                                                           | Musical Tension (p. 1)             |
| CC-49 | UN — Declaration on the Rights of Indigenous Peoples (A/RES/61/295)                                            | Full PDF               | <a href="https://www.un.org/development/desa/indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf">https://www.un.org/development/desa/indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf</a> | Historical Research (p. 1)         |
| CC-50 | "Laya (music)" — Wikipedia                                                                                     | Full text (thin)       | <a href="https://en.wikipedia.org/wiki/Laya_(music)">https://en.wikipedia.org/wiki/Laya_(music)</a>                                                                                                                                 | Loudness                           |

| ID    | SOURCE<br>(AUTHOR/VENUE<br>AS GIVEN) | RETRIEVED AS                                 | URL (AS GIVEN)                                                                                        | TIER |
|-------|--------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|------|
| CC-51 | "Pulse (music)"<br>— Wikipedia       | Full text                                    | <a href="https://en.wikipedia.org/wiki/Pulse_(music)">https://en.wikipedia.org/wiki/Pulse_(music)</a> | Low  |
| CC-52 | "Tempo rubato"<br>— Wikipedia        | <b>Search/snippet<br/>only —<br/>EXCLUDE</b> |                                                                                                       |      |

### 5.1.3 [ML-N] — LEARNING SCIENCE / RECOMMENDER ML (A3 §6; RETRIEVALS 2026-10-06)

| ID   | SOURCE (AUTHORS, YEAR<br>— TITLE — VENUE)                                                                                                                                                                                                             | RETRIEVED AS                                                                                                                                                                                               | TIER                                                    | CLAIM CLUSTER IT<br>SUPPORTS                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ML-1 | Manning, Raghavan & Schütze — Introduction to Information Retrieval, ch. 9 (Relevance feedback and query expansion) — full text:<br><a href="https://nlp.stanford.edu/IR-book/pdf/09expand.pdf">https://nlp.stanford.edu/IR-book/pdf/09expand.pdf</a> | full text                                                                                                                                                                                                  | Medium                                                  | Rocchio eq. 9.3; $\gamma < \beta$ convention; "tracks an evolving information need".                                                       |
| ML-2 | Hidasi, Karatzoglou, Baltrunas & Tikk (ICLR 2016) — Session-based Recommendations with RNNs — arXiv:1511.06939                                                                                                                                        | abstract + official repos via search                                                                                                                                                                       | Medium                                                  | GRU4Rec session lineage.                                                                                                                   |
| ML-3 | Brost, Mehrotra & Jehan (WWW 2019) — The Music Streaming Sessions Dataset — arXiv:1901.09851                                                                                                                                                          | abstract page                                                                                                                                                                                              | Medium                                                  | 160 M sessions; uniform-random subset for counterfactual evaluation.                                                                       |
| ML-4 | Garivier & Moulines (2008) — On UCB Policies for Non-Stationary Bandit Problems — arXiv:0805.3415                                                                                                                                                     | abstract                                                                                                                                                                                                   | Medium                                                  | Rewards constant over epochs, change at unknown instants; discounted/sliding-window UCB.                                                   |
| ML-5 | Li, Chu, Langford & Schapire (WWW 2010) — Contextual-Bandit Approach to Personalized News Article Recommendation — arXiv:1003.0146                                                                                                                    | search snippet                                                                                                                                                                                             | Medium                                                  | LinUCB.                                                                                                                                    |
| ML-6 | Russo et al. (2017) — A Tutorial on Thompson Sampling — arXiv:1707.02038                                                                                                                                                                              | search snippet                                                                                                                                                                                             | Medium                                                  | Thompson sampling.                                                                                                                         |
| ML-7 | Jannach, Lerche & Zanker (2018) — Recommending based on Implicit Feedback — Social Information Access chapter                                                                                                                                         | full text: <a href="https://web-ainf.aau.at/pub/jannach/files/BookChapter_Social_Information_Access_2018.pdf">https://web-ainf.aau.at/pub/jannach/files/BookChapter_Social_Information_Access_2018.pdf</a> | Medium; with [ML-23] ⇒ <b>High for "skip ≠ dislike"</b> | Verbatim: skips "can be context dependent and interpreting it as a general negative assessment of the previous track might be misleading". |
| ML-8 | Hu, Koren & Volinsky (ICDM 2008) — Collaborative Filtering for Implicit Feedback Datasets                                                                                                                                                             | text (chrisvolinsky.com PDF)                                                                                                                                                                               | Medium                                                  | (sign, magnitude, confidence); "vastly varying confidence levels".                                                                         |

| ID    | SOURCE (AUTHORS, YEAR — TITLE — VENUE)                                                                                | RETRIEVED AS                                                  | TIER       | CLAIM CLUSTER IT SUPPORTS                                                      |
|-------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------|--------------------------------------------------------------------------------|
| ML-9  | Chang, Zhang, Tang, Yin, Chang, Hasegawa-Johnson & Huang (WWW 2017) — Streaming Recommender Systems (sRec)            | retrieved snippets (experts.illinois.edu; archives.iw3c2.org) | Medium/Low | Time-decay / half-time framing.                                                |
| ML-10 | Carbonell & Goldstein (SIGIR 1998) — The Use of MMR, Diversity-Based Reranking — full text: cs.cmu.edu PDF            | full text                                                     | Medium     | $\lambda=1$ relevance / $\lambda=0$ diversity.                                 |
| ML-11 | Song, Kim, Lee et al. — Learning from Noisy Labels with Deep Neural Networks: A Survey — arXiv:2007.08199             | snippet                                                       | Medium     | Noisy-label learning.                                                          |
| ML-12 | Loni et al. (2021) — A Survey on Cold Start Problem in Recommender Systems — J. Big Data                              | snippet via references                                        | Medium/Low | Cold start.                                                                    |
| ML-13 | le et al. (2019) — RecSim — arXiv:1909.04847                                                                          | snippet                                                       | Medium     | Simulation environments for sequential recommendation.                         |
| ML-14 | Radlinski & Craswell (WSDM 2013) — Optimized Interleaving for Online Retrieval Evaluation                             | snippets (Microsoft/dblp/ACM)                                 | Medium/Low | Interleaving for online evaluation.                                            |
| ML-15 | Gama, Žliobaitė, Bifet, Pechenizkiy & Bouchachia (2014) — A Survey on Concept Drift Adaptation — ACM CSUR             | snippets + TU/e PDF                                           | Medium/Low | Concept drift.                                                                 |
| ML-16 | Gilotte, Calauzènes, Nedelec, Abraham & Dollé (2018) — Offline A/B Testing for Recommender Systems — arXiv:1801.07030 | abstract                                                      | Medium     | Counterfactual estimators; bias–variance limits.                               |
| ML-17 | Bottou, Peters, Quiñero-Candela et al. (JMLR 2013) — Counterfactual Reasoning and Learning Systems — arXiv:1209.2355  | abstract                                                      | Medium     | Counterfactual caution.                                                        |
| ML-18 | Nguyen, Hui, Harper, Terveen & Konstan (WWW 2014) — Exploring the Filter Bubble — full text: archives.iw3c2.org PDF   | full text                                                     | Medium     | Individual-level narrowing measured.                                           |
| ML-19 | Chen et al. (RecSys 2021) — Values of Exploration in Recommender Systems — ACM fullHtml, DOI 10.1145/3460231.3474236  | full text                                                     | Medium     | Verbatim: lack of exploration → satiation, "reduced enjoyment of the content". |
| ML-20 | Montecchio, Roy & Pachet (2019) — The Skipping Behavior of Users of Music Streaming Services — arXiv:1903.06008       | abstract                                                      | Medium     | Skip timing ↔ musical structure; stable under stationarity.                    |

| ID    | SOURCE (AUTHORS, YEAR — TITLE — VENUE)                                                                              | RETRIEVED AS | TIER                                   | CLAIM CLUSTER IT SUPPORTS                                          |
|-------|---------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------|--------------------------------------------------------------------|
| ML-21 | The universality of skipping behaviours on music streaming platforms — arXiv:2005.06987                             | snippet only | Low                                    | Skip profile as song-intrinsic.                                    |
| ML-22 | Google Analytics Help — About Analytics sessions                                                                    | snippet      | Medium (official doc; convention only) | 30-minute inactivity sessionization default.                       |
| ML-23 | Li, Kuo, Sheng, Zhang & Wu (CHI 2025) — Beyond Explicit and Implicit: How Users Provide Feedback — arXiv:2502.09869 | abstract     | Medium                                 | Intentional implicit feedback drives diversity/relevance purposes. |

5.1.4 INTERNAL (PROJECT) REFERENCES — NOT WORLD-EVIDENCE

| ID               | WHERE                                                                                                                                                        | ROLE                                                                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [SynAmp-dossier] | docs/research/synamp-brain-dossier.html                                                                                                                      | Internal dossier (metric taxonomy; feedback magnitudes; drift/exploration guidance). Used only for internal consistency.                                   |
| [SynAmp-CONTEXT] | .cluster/synamp-fast-brain/CONTEXT.md; plan schema read from apps/brain/src/query/plan.ts, signals.ts                                                        | Verified repo facts; registry gates; spot-check behaviours; scoping rules.                                                                                 |
| [INT-1]          | .cluster/synamp-fast-brain/CONTEXT.md                                                                                                                        | Signal registry & gates; spot-check behaviours (right/half/double/tap; 3-based ratios; "no steady beat"); epoch scoping; declared-vs-produced.             |
| [INT-2]          | docs/research/beat-timing-findings-2026-09-29.md                                                                                                             | Two-track smoke results; phase concentration $\approx 0.034$ with trackable pulse; abstention guards; "method rejection $\neq$ independent musical truth". |
| [INT-3]          | docs/synamp/plans/BEAT-TIMING.md                                                                                                                             | Onset-attribution requirements; "do not repeat" list (incl. not calling rejected tracks arrhythmic).                                                       |
| [INT-4]          | docs/research/synamp-brain-dossier.html                                                                                                                      | "BPM is the weakest signal"; octave errors 60/120/240; proxy field; dataset-bias design row; cross-cultural similarity variance (Papaioannou).             |
| [R-code]         | apps/brain/src/session/{events,session,feedback}.ts; apps/brain/src/query/{evaluate,signals,plan}.ts; docs/synamp/plans/AGENT-ROADMAP.md (P3); dossier ch. 7 | Internal design source (read-only, retrieved 2026-10-06).                                                                                                  |

5.1.5 UNRETRIEVED / BLOCKED IN THIS PASS (DO NOT CITE AS EVIDENCE)

*Union of the A1 / A2 / A3 unretrieved lists, as given by each pack. Nothing below was fetched; none of it may be cited as evidence in the report. Shown here so the boundary of what the evidence actually covers stays visible.*

**From A1 (compiled from its §5/§6 annotations + return block; reasons as given):**

1. Karageorghis & Priest — review **Part I** — UNRETRIEVED (PMC hit was a different paper; Part II links its content).
2. [MP-7] Heiderscheit & Madson 2015 — UNRETRIEVED (publisher/blocks; record metadata verified via search + Semantic Scholar). Cluster covered by [MP-8][MP-9].
3. [MP-11] Janata, Tomic & Haberman 2012 — UNRETRIEVED (closed; metadata verified via Semantic Scholar API).
4. [MP-12] Stupacher, Hove & Janata 2016 — UNRETRIEVED (closed; record verified via search).
5. [MP-15] Sloboda 1991 — UNRETRIEVED (paywall; mirrors blocked).
6. [MP-23] Husain et al. 2002 — UNRETRIEVED (SAGE blocked).

7. [MP-24] Pietschnig et al. 2010 — UNRETRIEVED (ScienceDirect captcha). Cluster covered by [MP-25].
8. Zillmann 1988 (American Behavioral Scientist) — UNRETRIEVED (noted under [MP-6]; mood-management primary).
9. van Goethem & Sloboda 2011 — UNRETRIEVED (listed in A1 return-block gaps; no further detail given).
10. de la Mora Velasco 2023 — UNRETRIEVED here (dossier only; conflict #1).
11. Kiss & Linnell 2021 (mind-wandering suppression) — UNRETRIEVED (described in [SynAmp-dossier]; candidate).
12. ISE origins: Colle & Welsh 1976; Salamé & Baddeley 1982 — originals not retrieved (cited within [MP-20]).
13. Binaural-beats evidence: Ingendoh et al. 2023 vs Garcia-Argibay — UNRETRIEVED here (conflicting evidence; must not be implemented as science-backed).
14. Driving-simulator companion work — dossier only; UNVERIFIED here (A1 §1.7).
15. Coverage-check note (as given): "Mood management (Zillmann) — Gap (primary UNRETRIEVED; secondary only)".

**From A2 — explicit UNRETRIEVED list (verbatim):**

1. Clayton, *Time in Indian Music* (SOAS repository page found but not fetched content) — India analysis relies on [CC-6]/[CC-38] only.
2. Locke, *Drum Gahu* (book; only pointer via [CC-2]) — Akan/Ewe detail beyond [CC-1, CC-2] not verified.
3. Agawu 2006 full text (PDF fetch empty).
4. Fürniss 2006 full text (Anubis block).
5. Saraga paper text (emusicology.org Anubis block); Saraga README fetched but unread — no claims made.
6. Polak 2018 full text (KTH DiVA Anubis block; abstract used).
7. Jacoby & McDermott 2017 full text (cell.com Cloudflare block; abstract used).
8. Surjodiningrat 1972 original tuning measurements (via [CC-11] only).
9. Indian intonation MIR (tonic identification / pitch-tracking papers; UPF repository blocked).
10. "U-M CSEAS Spirit of Tuning" article (Cloudflare block).
11. NMA "Songlines" exhibition page (JS-only shell).
12. Direct experimental evidence for "note-density → perceived speed/tempo bias" (searches unsuccessful; claim kept candidate, §3.5).
13. AIATSIS materials (could not retrieve usable page this pass; do not cite AIATSIS as a source).
14. GlobalMood / "Missing Melodies" / music-AI fairness papers surfaced in search but not retrieved — excluded entirely from citations.

**From A3 — UNRETRIEVED / partially unretrieved (verbatim):**

Rocchio (1971) original paper; Ide dec-hi original; WSDM-2019 challenge winning-solution details; Seshadri et al. RecSys'23 (contrastive skip learning — dossier citation only); "Why People Skip Music?" (2023) full text (snippets only); Xie et al. WWW'23 dwell reweighting full text (dossier names arXiv:2209.09000 — not independently retrieved); Spotify's "first 30 seconds" rule (dossier already labels unverified — do not use as a finding). The dossier's internal magnitudes/claims remain project-internal candidate values, not independently validated here.

## 5.2 Parameter table P1–P25 — A3 §3.7 spec, annotated with implementation (B2 §3) and review outcomes (C1)

Columns: (#) parameter · default → implemented as (B2 §3 names/constants, code) → gate / review note (C1 probe results; deviations and judgment calls worth knowing when reading Ch.2).

| #   | PARAMETER · DEFAULT<br>(A3 §3.7)                        | IMPLEMENTED AS (B2 §3)                                                                                                                     | GATE / REVIEW NOTE (C1 + B2 DEVIATIONS)                                                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1  | <code>G_idle</code> idle gap = 30 min                   | <code>IDLE_GAP_MS = 30 * 60_000</code> (epochs.ts); mask, not decay                                                                        | C1: 30:00 exact keeps / 30:01 splits (value 0); missing <code>session_id</code> = its own bucket (sandwich → 3 epochs). ✓                                                                                                                                                                                                                                                        |
| P2  | Day boundary = local midnight                           | <code>resolveEpochs</code> day-boundary; injectable IANA timezone (default server local)                                                   | C1: midnight split with 60 s gap ✓; pin timezone for cross-restart day determinism (B2 §6.8).                                                                                                                                                                                                                                                                                    |
| P3  | <code>h_E</code> epoch half-life = 30 min               | <code>EPOCH_HALF_LIFE_MS</code> ; <code>0.5^(Δt/30min)</code>                                                                              | C1: ratio exactly 0.25 at +60 min ✓ (" +60 min ≤ 0.25× fresh" asserted on derived values).                                                                                                                                                                                                                                                                                       |
| P4  | <code>G_live</code> active window = <code>G_idle</code> | <code>LIVE_WINDOW_MS</code> ; <code>activeEpoch</code>                                                                                     | C1: expiry live at exactly +30:00, cleared at +30:00.001 with "no active session" note ✓. Deviation (B2 §5.2): epochs ending after <code>now</code> are never "active".                                                                                                                                                                                                          |
| P5  | <code>W</code> bound = 0.15                             | evaluate.ts <code>0.15 · tanh(v/2)</code> (unchanged); one saturation                                                                      | C1: flood 20 loves/skips → 0.15000 / −0.14925 / 0.15000, all parts finite; hard rules inviolable under maxed bonus. ✓                                                                                                                                                                                                                                                            |
| P6  | tanh scale = 2                                          | same ( <code>0.15 · tanh(v/2)</code> )                                                                                                     | C1: monotonicity + saturation spot-checks ✓; verified as composed through the real pipeline, not raw value.                                                                                                                                                                                                                                                                      |
| P7  | love = +2.0 / c=1.0                                     | <code>EPOCH_EVIDENCE</code> + <code>PERSISTENT_EVIDENCE</code> (v1 parity)                                                                 | C1: single-love bonus = <b>0.11424</b> (brief's "0.10–0.11" was approximate); part "you loved this" present. ✓                                                                                                                                                                                                                                                                   |
| P8  | thumb_up = +1.0 / c=0.9                                 | as above                                                                                                                                   | C1: explicit-only parity with <code>deriveFeedback</code> (deep-equal adjust/removed) ✓.                                                                                                                                                                                                                                                                                         |
| P9  | thumb_down = −1.0 / c=0.9                               | as above                                                                                                                                   | Persists via L1 when reason ≠ <code>not_now</code> ; <code>remove + reason=not_now</code> → epoch hide only (C1). ✓                                                                                                                                                                                                                                                              |
| P10 | repeat = +1.0 / c=0.7                                   | <code>EPOCH_EVIDENCE</code>                                                                                                                | L2 only; dies after epoch end ✓ (constant-table check).                                                                                                                                                                                                                                                                                                                          |
| P11 | full_play = +0.5 / c=0.5                                | <code>EPOCH_EVIDENCE</code>                                                                                                                | L2 only (constant-table check; low confidence = unintentional signal).                                                                                                                                                                                                                                                                                                           |
| P12 | skip_early = −0.6 / c=0.5                               | <code>EPOCH_EVIDENCE</code>                                                                                                                | C1: k=1 → −0.2999, no hide; k=2 → hide + −0.5897 ✓. "Skip ≠ dislike" (High with [ML-7]+[ML-23]).                                                                                                                                                                                                                                                                                 |
| P13 | skip_late = −0.1 / c=0.2                                | <code>EPOCH_EVIDENCE</code>                                                                                                                | Never hides ✓ (constant-table check).                                                                                                                                                                                                                                                                                                                                            |
| P14 | external_play = +0.25 / c=0.5                           | <code>EPOCH_EVIDENCE</code> (v1 value kept)                                                                                                | L2 only; proposal at ≥5 epochs / ≥3 dates ✓ (P22 probe). Demotion decision noted in A3 gaps.                                                                                                                                                                                                                                                                                     |
| P15 | skip-hide count = 2                                     | Epoch-wide hide from <code>skip_early ×2</code> (B2 §5.6: v1's playlist-scoped rule removed from L1)                                       | C1: full hide lifecycle ✓ incl. exact +30:00 expiry edge; hide survives re-resolve within epoch. Judgment flagged to C-round: epoch-wide vs playlist-scoped scope.                                                                                                                                                                                                               |
| P16 | <code>not_now</code> = hide n=1, no L1                  | derive reset/not_now semantics                                                                                                             | C1: <code>not_now</code> hides at n=1; value 0 after the epoch; no persistent cell appears ✓; reset note present.                                                                                                                                                                                                                                                                |
| P17 | artist π=0.35 / cap=0.8 / min-ev ≥2                     | <code>ARTIST_MIN_EVENTS=2</code> , <code>ARTIST_MIN_SUM=1.0</code> ; cap ±0.8; <code>π · Σ</code>                                          | C1 <b>F2 (wording drift)</b> : A3 §3.3(b) prose says "≥2 <b>distinct tracks</b> "; code + P17 table + dispatch say "≥2 <b>events</b> " — <b>code follows P17; A3 prose is the outlier</b> (fix the prose or accept; behaviour per table). C1: cap +0.8 exactly; single event ≤0.11; entity-only propagation (loves don't feed the artist term — state this in one line in Ch.2). |
| P18 | centroid gates n_K≥3, cov≥0.6, n_R≥2                    | <code>FEATURE {min_keeps:3, min_rejects:2, coverage:0.6}</code> ; plus axis needs ≥4 usable library values + nonzero 1.4826-MAD (B2 §5.11) | C1: coverage opens at exactly 60%, falls back at 40 % with exact note; n=2 kept ⇒ no axis + reliability note ✓.                                                                                                                                                                                                                                                                  |
| P19 | β/γ = 0.5/0.15                                          | <code>FEATURE {beta:0.5, gamma:0.15}</code> ; γ<β enforced                                                                                 | C1: γ<β enforced; total clipped to exactly 1.0 (sum-of-parts 2.35) ✓.                                                                                                                                                                                                                                                                                                            |
| P20 | feature caps: axis ±0.5 / total ±1.0                    | per-axis parts ≤0.5 by construction; total clip ±1.0; artist clip ±0.8                                                                     | C1: per-axis 0.47 ≤ 0.5; adversarial wide-spread set shrunk via 1/(1+s²) ✓.                                                                                                                                                                                                                                                                                                      |
| P21 | neg proposal ≥3 epochs ^ ≥2 dates                       | <code>PROPOSAL_THRESHOLDS</code>                                                                                                           | C1: 2→none; 3/2 dates→exactly 1 with <code>{epochs:3,dates:2,dayparts:[morning,afternoon]}</code> ; same-date→none ✓. Note: the <code>not_now_pattern</code> "≥2 dayparts" gate                                                                                                                                                                                                  |

| #   | PARAMETER · DEFAULT<br>(A3 §3.7)       | IMPLEMENTED AS (B2 §3)                                           | GATE / REVIEW NOTE (C1 + B2 DEVIATIONS)                                                                                                                                          |
|-----|----------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                        |                                                                  | verified by code read + constants only (C1 §8.2); live epoch excluded from scan.                                                                                                 |
| P22 | pos proposal ≥5 epochs<br>^ ≥3 dates   | PROPOSAL_THRESHOLDS (repeat 4/3, external 5/3)                   | C1: 5/3→proposal; 4→none ✓.                                                                                                                                                      |
| P23 | proposal scan window = 30 d            | PROPOSAL_SCAN_DAYS = 30                                          | C1: 35-day-old pattern → none ✓.                                                                                                                                                 |
| P24 | exploration OFF; if ON: 1 slot, seeded | explorationEnabled / pickExplorationSlot (derive never reads it) | C1: OFF byte-identical to no-budget; ON never alters the derived view; seeded slot deterministic ✓.                                                                              |
| P25 | scope_persistence = declared           | PolicyOptions.scopePersistence default "declared"                | C1: both modes ✓. global_only magnitudes (±1.0 / c=0.9 / epoch half-life; labels "... (this session)") are a B2 §5.5 judgment call — one constant to change if review disagrees. |

**Reading the table (footnotes):** – C1 re-derived ≥10 rows plus the full constant tables ( EPOCH\_EVIDENCE , PERSISTENT\_EVIDENCE , FEATURE , PROPOSAL\_THRESHOLDS ); everything above "reproduced green" except the noted wording drift (F2) and the two judgment calls (P15 scope, P25 magnitudes) which are flagged, not smoothed. (C1 §2 F5 also notes cosmetic label drift for the artist part: quote the implemented label "other tracks by this session", not A3's example wording.) – C1's one **major must-fix (F1, wave arc on odd measured counts → undefined /500) is outside the P-table** (it is in sequencing). It was fixed by H1 (interleave bound + guards; n=1/3/5 regression tests; resolve path re-probed OK). Post-fix C1 verdict expected: pass-with-notes. – "L1 explicit cells survive a learning reset" (B2 §5.4) is implemented as documented and judged acceptable by C1 §7; if the report wants "forget" to also suppress pre-reset explicit cells, that is a one-spot change — say so explicitly. – Counts in this table's sources: A3 §3.7 (spec), B2 §3 (implementation), C1 §3–§4 (review probes). All fixtures are synthetic; nothing here is evidence about real music.

## 5.3 Glossary (16 terms, plain language)

| TERM                                | PLAIN-LANGUAGE MEANING (AS IMPLEMENTED HERE)                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>plan</b>                         | The machine-readable description of what the user asked for: constraints + sequencing + declared relaxation rules; versioned schema (v2.0), validated, and hashed (the plan hash). Something the brain <i>can</i> enforce — vague asks produce questions, impossible asks are refused honestly.                                                          |
| <b>constraint (hard / soft)</b>     | One requirement inside a plan (e.g., "bpm ≤ 120"). <b>Hard</b> = a guard that is never auto-relaxed (explicit exclusions must be hard). <b>Soft</b> = may only be relaxed through the plan's declared ladder ( drop_boost / widen_numeric ).                                                                                                             |
| <b>unknown_policy</b>               | What to do when a track's value for a constraint is unknown: exclude / include / neutral . Declared-but-no-producer fields use neutral + the inert note "no producer yet — inert on real data".                                                                                                                                                          |
| <b>near-miss</b>                    | The second, separately-labelled result tier: tracks that miss strict filtering within registered near-miss widths. Near-misses are shown, never receive learning bonuses, and never carry feedback reasons.                                                                                                                                              |
| <b>tier</b>                         | Which bucket a track lands in after evaluation: <b>strict</b> (passes every constraint — the only place learning/scoring applies) vs <b>near-miss</b> (close misses, hand-labelled). Not to be confused with source-confidence tiers in §5.1.                                                                                                            |
| <b>arc / sequencing</b>             | The ordering step applied <i>after</i> evaluation to the strict tier only (a permutation — nothing added or dropped; unknowns keep their places). Arcs: flat / build / peak / cooldown / wave ; notes appear in sequencing_applied (e.g., "arc build: ordered by measured pace/energy proxy...").                                                        |
| <b>epoch</b>                        | The learning window: one listening session, split by session change, a 30-minute idle gap, or local midnight. Implicit learning <b>never crosses an epoch boundary</b> — a closed epoch contributes exactly nothing. Missing session_id events form their own bucket.                                                                                    |
| <b>daypart</b>                      | Coarse time-of-day label attached to an epoch (morning 05–12 / afternoon 12–17 / evening 17–22 / night 22–05, from the epoch start). A display/evidence convention only — no empirical basis claimed.                                                                                                                                                    |
| <b>hide (persistent vs session)</b> | A suppressed track. <b>Persistent remove</b> = playlist-scoped, lasts until restored ( hidden_by_you ). <b>Session hide</b> = epoch-wide, from skip_early ×2 or not_now ; clears when the session ends (or via "Forget this session"; hidden_by_session ). Both leave the strict tier; the two counts are reported separately (no dead Restore buttons). |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>proposal</b>              | A cross-epoch suggestion — the <i>only</i> channel through which learning crosses epochs. Backed by evidence { <code>epochs</code> , <code>dates</code> , <code>dayparts</code> } and conservative thresholds; requires user confirmation before anything changes. Accepting appends a normal global thumb event (track subjects) or records the decision (artist subjects); never an automatic playback change. |
| <b>evidence weight</b>       | How strongly one event counts: magnitude × confidence (e.g., love +2.0 @ c=1.0; repeat +1.0 @ c=0.7), then decayed by the epoch half-life. These are the numbers the learning is built from — all candidate values, tunable, versioned with the policy.                                                                                                                                                          |
| <b>reliability gate</b>      | The filter deciding whether measured data is trustworthy enough to learn from: declared/missing fields are refused; <code>bpm</code> uses <code>tempo_confidence</code> (full ≥0.5, damped 0.35, unusable ≤0.2/missing); a centroid axis additionally needs ≥4 usable library values and a non-zero robust scale. Gated data produces honest notes, never zero-substitution.                                     |
| <b>centroid</b>              | The evidence "centre of mass" per feature axis (e.g., bpm) of the tracks you kept / skipped this session; candidates near the kept centroid get a small, bounded pull ("tempo ~126 BPM — you kept three such tracks"). Gated by minimum counts and coverage; per-axis pull capped.                                                                                                                               |
| <b>saturation bound</b>      | The total feedback adjustment is squeezed through $0.15 \cdot \tanh(v/2)$ — <b>never more than ±0.15</b> , applied once, inside the strict tier only. It is a tiebreaker, never an override: hard constraints and explicit exclusions stay inviolable no matter how much evidence accumulates.                                                                                                                   |
| <b>policy version</b>        | The label stamped on events and readouts for <i>which meaning of the numbers</i> produced them ( <code>heuristic-v1</code> or <code>epoch-v1</code> ). Any meaning change = new version + re-derive from the same immutable event log — never a migration. Rollback = switch <code>listening_policy</code> back ( <code>legacy-v1</code> ).                                                                      |
| <b>spot-check correction</b> | The human-in-the-loop tempo fix: you verify right / half / double / tap, or "no steady beat" (3-based metre ratios recognised). Corrections are applied only while the measured BPM matches the checked one; this is the seed of a human-verified reliability tier.                                                                                                                                              |

## 5.4 Reproduction commands — full verification stack

Consolidated from B4b §"Exact commands", B5 §3.2, C1 §10, H1 §4, H2 "Verification" and the H4 re-sync (receipt `H4-lab-sync.md`; canonical lab capture `evidence/final-lab/`). S5's delivery-manifest re-capture remains the definitive set. Captures were taken on `feat/fast-learning-brain` @ base `b6ea5ce1` + the working tree; no commits at capture time, no installs.

```
CLUSTER=/Users/tapps/.openclaw-autoclaw/agents/algorithm-scientist/workspace/.cluster/synamp-fast-brain
cd /Users/tapps/_dev/web-apps/SynAmp

# 1) Brain test suite – expect the ONE carried red (macOS librarian case-fold) + /dev/shm skip
pnpm --filter @synamp/brain test

# capture (H1, 00:02 CDT): 218 tests · 216 pass · 1 fail · 1 skip · exit 1 by design – reproduced fresh post-hotfix
# (earlier captures: H2 212/210, C1 210/208 – tests landed progressively; the S5 re-run is definitive)

# 2) Type-checks – both exit 0. Root `pnpm type-check` stops in vendored webamp (pre-broken, untouched): expect exit 1 there
pnpm --filter @synamp/brain type-check
pnpm --filter @synamp/web type-check
pnpm --filter @synamp/web build # exit 0 ("✓ built in ~1s")

# 3) Verification harness (deterministic; re-run before quoting any numbers)
node --experimental-strip-types tools/brain-lab/lab.mts

# expected: 8 scenarios · 86 checks pass · 0 fail · 0 pending · 5 skip · exit 0
# (last canonical green = H4 re-sync, post-hotfix – `evidence/final-lab/`; a mid-round transient came from concurrent evaluate)

# 4) Route smoke – self-contained scripts; scratch ports/dirs only; NEVER real data
bash $CLUSTER/evidence/b5/route-smoke-epoch.sh # epoch mode + forget + legacy switch
bash $CLUSTER/evidence/b5/route-smoke-auth.sh # PLAYLIST_API_TOKEN 401/200
bash $CLUSTER/evidence/b5/route-smoke-resolve.sh # resolve == explain == evaluate; hide → re-derive
# transcripts: $CLUSTER/evidence/b5/route-smoke-*.log · H2 variants: evidence/h2/smoke1-routes.mjs, smoke2-playlist-explain.mj

# 5) Independent adversarial probes (C1; ~132 checks) – read-only wrt the repo
```

```
bash $CLUSTER/evidence/c1/runall.sh

# C1-era results: p1 20/0 · p2 41/0 · p3 37/1 (the 1 = the wave defect, fixed by H1) · p4 33/0 · p5 now OK

# *p3 re-runs may show cosmetic fails: its hard-coded note strings predate the F8 wording fix (H1 §5.4); behaviour is what cou
```

**Carried reds to expect (unchanged by this branch):** (1) macOS librarian case-fold test

( `librarian.test.ts:149` ) — fails on default APFS case-insensitive filesystems, also red on `main` ;

(2) root `pnpm type-check` chain — 11 pre-existing `packages/webamp` tsc errors; (3) `/dev/shm`

cross-disk skip. Full accounting: C1 §5, B4 §A, H2.

**When quoting numbers in the report:** use the S5 delivery-manifest re-capture, reproducing the

H4/C4 runs (brain suite, type-checks, brain-lab JSON, route smoke transcripts, `evidence/final-`

`lab/hashees.txt` for module hashes). Do **not** quote B4's 64-check lab number, B6/roadmap's

"Brain 205", or the stale `interpret.ts` hash `f260b260...` — all superseded (C1 §6 F4).

---

*End of Ch.5 appendix. Sources: retrieved by A1/A2/A3 on 2026-10-06 (modes as stated per entry); this merge adds no sources and performs no retrieval. Unretrieved items are listed in §5.1.5 and must not be cited as evidence.*

**How to verify this dossier.** Every claim carries its source ID ([MP-n] / [CC-n] / [ML-n] point into the appendix) or names the receipt it came from. The full stack re-runs with: `pnpm --filter @synamp/brain test`, `pnpm --filter @synamp/brain type-check && pnpm --filter @synamp/web type-check`, and `node --experimental-strip-types tools/brain-lab/lab.mts`. Logs, screenshots and checksums live in the delivery's `verification/` folder; the manifest there is the frozen record this document was assembled against.

**Colophon.** Assembled by the SynAmp agent cluster from research packs A1-A4, engineering receipts B1-B6/H1-H5, and independent reviews C1-C4; rendered as a single self-contained HTML file (no external assets, no scripts). Counts and checksums are frozen in the delivery manifest under `verification/`.